

Mathematics A

Unit 2 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

81 classified items • 317 marks • 18 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

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Unit 2 • Expertise • Unit 2 • Expertise Q16+ • 81 items • 317 marks

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June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Direct & Inverse Proportion

Unit 2 • Expertise • 19 marks • 5 item(s)

June 2025 | 4MA1-1H Q18 | 4 marks

Recover the input under inverse variation

June 2023 | 4MA1-2H Q16 | 5 marks

Root direct variation $y = k\sqrt{x}$

June 2022 | 4MA1-2H Q17 | 4 marks

Power direct variation $y = kx^n$

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

June 2021 | 4MA1-2H Q16 | 3 marks

Inverse-square or inverse-power variation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebra Toolkit

Unit 2 • Expertise • 7 marks • 2 item(s)

June 2024 | 4MA1-2H Q20 | 3 marks

Signed, decimal or fractional formula substitution

June 2023 | 4MA1-2H Q17 | 4 marks

Subject appears in two linear terms

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Completing the Square

Unit 2 • Expertise • 3 marks • 2 item(s)

November 2021 | 4MA1-1H Q19 | 2 marks

Negative leading coefficient or maximum form

November 2021 | 4MA1-1H Q19 | 1 marks

Negative leading coefficient or maximum form

Original question context and synchronized companion pages follow.

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Rearranging Formulas

Unit 2 • Expertise • 8 marks • 2 item(s)

June 2025 | 4MA1-2H Q16 | 4 marks

Single squared/powered subject

January 2020 | 4MA1-2H Q16 | 4 marks

Subject in the denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Algebraic Proof

Unit 2 • Expertise • 16 marks • 5 item(s)

January 2022 | 4MA1-2H Q18 | 3 marks

Fixed remainder modulo a divisor

January 2020 | 4MA1-2H Q17 | 3 marks

Consecutive integer parity identity

June 2021 | 4MA1-1H Q17 | 3 marks

Consecutive even/odd sequence proof

November 2021 | 4MA1-1H Q19 | 3 marks

Derive a general geometric algebra formula

January 2023 | 2H Q17 | 4 marks

Product or sum of arbitrary odd/even numbers

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Solving Inequalities

Unit 2 • Expertise • 16 marks • 4 item(s)

June 2025 | 4MA1-2H Q19 | 3 marks

Factorable quadratic sign intervals

January 2022 | 4MA1-2H Q22 | 5 marks

Contextual quadratic feasible range

May-Nov 2020 | 2H Q17 | 5 marks

Inequality symbol from a number line

January 2023 | 2H Q22 | 3 marks

One-sided linear inequality

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Simultaneous Equations

Unit 2 • Expertise • 25 marks • 5 item(s)

June 2024 | 4MA1-2H Q22 | 5 marks

Line with mixed quadratic relation

November 2024 | 4MA1-2H Q22 | 5 marks

Line with circle or sum-of-squares curve

January 2020 | 4MA1-2H Q22 | 5 marks

Multiple solution-pair validation

May-Nov 2020 | 2H Q16 | 5 marks

Draw the missing line for an intersection

January 2023 | 2H Q20 | 5 marks

Draw the missing line for an intersection

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Sequences

Unit 2 • Expertise • 25 marks • 5 item(s)

June 2024 | 4MA1-1H Q24 | 6 marks

Nonlinear or integer-constrained series recovery

November 2023 | 4MA1-1H Q19 | 6 marks

Recover number of arithmetic terms

November 2024 | 4MA1-1H Q19 | 5 marks

Recover first term and common difference

January 2021 | 4MA1-1H Q24 | 4 marks

Symbolic relation between partial sums

May-Nov 2020 | 1H Q24 | 4 marks

Arithmetic n th term from listed terms

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Functions

Unit 2 • Expertise • 19 marks • 7 item(s)

June 2025 | 4MA1-1H Q16 | 1 marks

Evaluate a function at a numerical input

June 2025 | 4MA1-1H Q16 | 3 marks

Solve a composite function inequality

June 2023 | 4MA1-1H Q17 | 4 marks

Exclude a denominator zero from the domain

January 2022 | 4MA1-2H Q25 | 1 marks

Domain and range swap under inversion

June 2021 | 4MA1-1H Q24 | 2 marks

Evaluate a numerical composite function

June 2021 | 4MA1-1H Q24 | 4 marks

Inverse of a restricted quadratic function

January 2023 | 1H Q18 | 4 marks

Inverse of a composite-defined function

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Graphs of Functions

Unit 2 • Expertise • 10 marks • 3 item(s)

November 2024 | 4MA1-1H Q23 | 5 marks

Infer a graph parameter from two intersections

January 2022 | 4MA1-2H Q21 | 2 marks

Estimate roots from axis crossings

January 2022 | 4MA1-2H Q21 | 3 marks

Draw a line to solve a nonlinear equation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Transformations of Graphs

Unit 2 • Expertise • 11 marks • 8 item(s)

June 2024 | 4MA1-2H Q25 | 2 marks

Horizontal translation

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical scale or x-axis reflection

November 2024 | 4MA1-2H Q19 | 1 marks

Vertical translation

June 2022 | 4MA1-2H Q23 | 1 marks

Horizontal scale or y-axis reflection

November 2021 | 4MA1-2H Q21 | 1 marks

Transform an explicit polynomial and simplify

November 2021 | 4MA1-2H Q21 | 1 marks

Translation vector applied to $y=f(x)$

November 2021 | 4MA1-2H Q21 | 1 marks

Describe from two curve equations

November 2021 | 4MA1-2H Q21 | 3 marks

Combined feature mapping

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Differentiation

Unit 2 • Expertise • 34 marks • 7 item(s)

June 2025 | 4MA1-2H Q17 | 6 marks

Coordinates of polynomial stationary points

November 2023 | 4MA1-2H Q18 | 4 marks

Velocity and acceleration from displacement

November 2024 | 4MA1-1H Q20 | 4 marks

Equation of a calculus tangent

January 2022 | 4MA1-1H Q24 | 6 marks

Recover curve parameters from point and gradient data

January 2022 | 4MA1-2H Q24 | 5 marks

Instantaneous rest or direction reversal

May-Nov 2020 | 1H Q23 | 4 marks

Points where gradient equals a value

January 2023 | 1H Q20 | 5 marks

Maximum volume from geometric dimensions

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Angles in Polygons & Parallel Lines

Unit 2 • Expertise • 6 marks • 1 item(s)

June 2022 | 4MA1-1H Q23 | 6 marks

Solve an irregular polygon angle or side-count problem

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Bearings, Scale Drawing & Constructions

Unit 2 • Expertise • 16 marks • 3 item(s)

November 2023 | 4MA1-1H Q18 | 5 marks

Route geometry followed by distance/time synthesis

November 2024 | 4MA1-2H Q24 | 6 marks

Sine or cosine rule supports a final bearing

January 2021 | 4MA1-1H Q16 | 5 marks

Sine or cosine rule supports a final bearing

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Circle Theorems

Unit 2 • Expertise • 14 marks • 3 item(s)

November 2023 | 4MA1-1H Q22 | 6 marks

Use radius perpendicular to tangent

November 2024 | 4MA1-2H Q16 | 5 marks

Complete a multi-theorem circle-angle chain

January 2022 | 4MA1-2H Q19 | 3 marks

Use the internal intersecting-chords product

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Volume & Surface Area

Unit 2 • Expertise • 28 marks • 6 item(s)

June 2024 | 4MA1-1H Q23 | 5 marks

Find volume or surface after removing a solid

November 2023 | 4MA1-1H Q21 | 5 marks

Find volume of a prism from its cross-section

June 2022 | 4MA1-1H Q22 | 5 marks

Find volume of a joined composite solid

January 2020 | 4MA1-2H Q26 | 6 marks

Find cone surface area or recover its sector

January 2021 | 4MA1-2H Q17 | 4 marks

Find surface area of a cuboid or simple prism

January 2023 | 2H Q21 | 3 marks

Find sphere or hemisphere surface area

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Area & Volume of Similar Shapes

Unit 2 • Expertise • 22 marks • 5 item(s)

June 2025 | 4MA1-2H Q26 | 4 marks

Solve a volume difference or sum for similar solids

November 2023 | 4MA1-1H Q23 | 5 marks

Convert between area and volume ratios

November 2023 | 4MA1-2H Q23 | 5 marks

Use similar solids to solve a frustum problem

January 2022 | 4MA1-1H Q17 | 4 marks

Solve a volume difference or sum for similar solids

January 2023 | 2H Q19 | 4 marks

Solve an area difference or sum for similar figures

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Vectors

Unit 2 • Expertise • 38 marks • 8 item(s)

June 2025 | 4MA1-1H Q23 | 5 marks

Find an intersection vector from two paths

June 2024 | 4MA1-2H Q24 | 6 marks

Find an intersection vector from two paths

November 2023 | 4MA1-2H Q22 | 5 marks

Recover a division ratio using vector parameters

June 2023 | 4MA1-1H Q19 | 5 marks

Recover a division ratio using vector parameters

November 2024 | 4MA1-1H Q24 | 6 marks

Find an intersection vector from two paths

November 2024 | 4MA1-2H Q17 | 3 marks

Find magnitude of a vector

May-Nov 2020 | 2H Q19 | 3 marks

Use vectors in a parallelogram, trapezium or polygon path

January 2023 | 1H Q24 | 5 marks

Find a section or midpoint vector

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

WORKING SPACE

Inverse proportionality

4 marks

Find the constant

$$y = \frac{k}{x^3}, \quad 4 = \frac{k}{3^3} \Rightarrow k = 108.$$

Use y=864

$$864 = \frac{108}{x^3} \Rightarrow x^3 = \frac{1}{8} \Rightarrow x = \frac{1}{2}.$$

Final answer

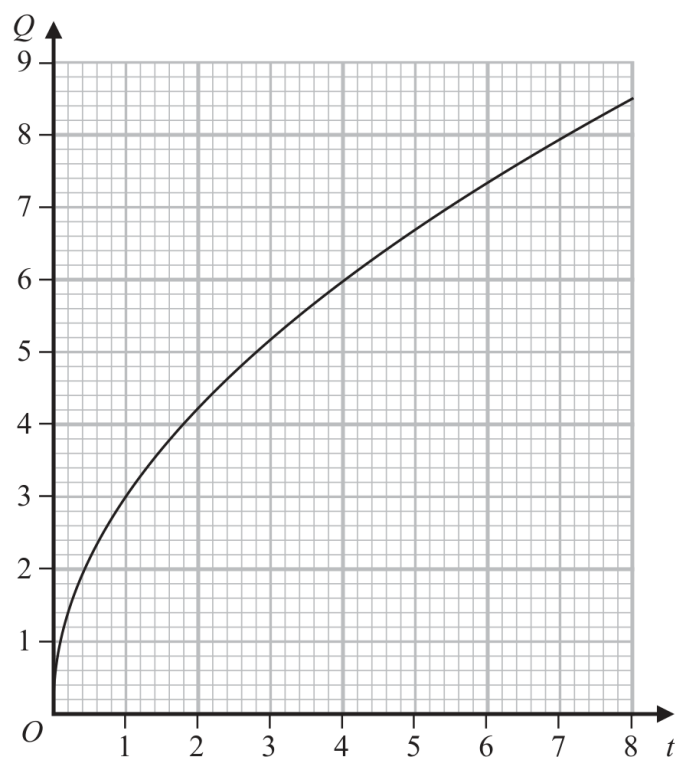
$$x = \frac{1}{2}$$

4MA1-2H Q16

5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

Continue your method

5 marks

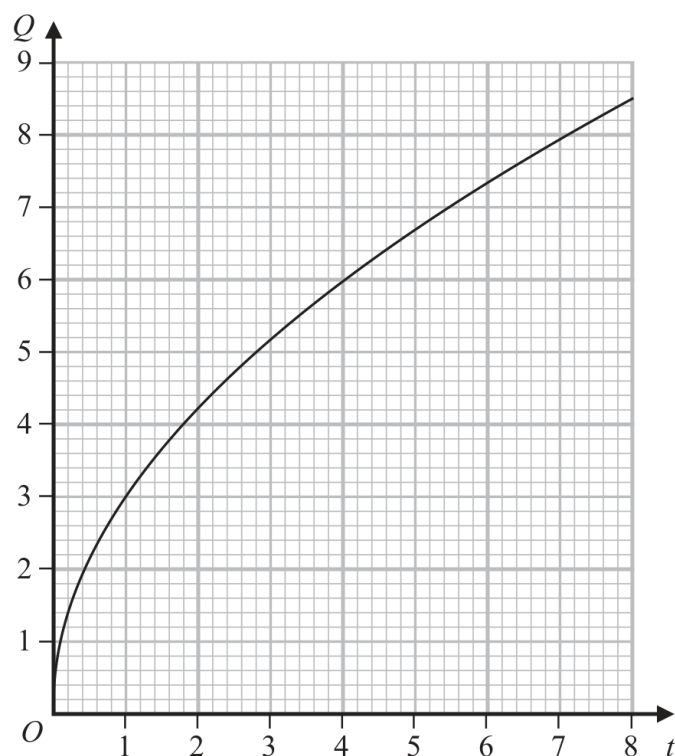
WORKING SPACE

4MA1-2H Q16

5 marks

16 Q is directly proportional to $\sqrt{t}$

The graph shows the relationship between Q and t for $0 < t < 8$



(a) Find a formula for Q in terms of t

.....
(3)

Q is increased by 20%

(b) Find the percentage increase in t

..... %
(2)

Root proportionality and percentage change

5 marks

Read the graph

$$Q \propto \sqrt{t}, \quad Q = k\sqrt{t}; \text{ a point such as } (1, 3) \text{ gives } k = 3.$$

Write the model

$$Q = 3\sqrt{t}.$$

Apply the 20% change

$$Q' = 1.2Q \Rightarrow 3\sqrt{t'} = 1.2(3\sqrt{t}) \Rightarrow t' = 1.2^2 t = 1.44t.$$

Convert to a percentage

$$(1.44 - 1) \times 100\% = 44\%.$$

Final answer

$$Q = 3\sqrt{t}; \quad 44\%$$

4MA1-2H Q17 Q17

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

WORKING SPACE

4MA1-2H Q17 Q17

4 marks

- 17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

Chapter: Direct & Inverse Proportion

4 marks

Q17 – Chapter: Direct & Inverse Proportion

Family: Build an algebraic direct-variation model • Variant: Power direct variation $y=kx^n$ **Method**Direct variation gives $M = kh^3$.*Why: Translate the variation statement into an equation.***Method** $4 = k(0.5)^3$, so $k = 32$.*Why: Use the given pair to determine the constant.***Method** $500 = 32h^3$, so $h^3 = 15.625$.*Why: Substitute the new M value and isolate h cubed.***Method** $h = \text{cube root of } 15.625 = 2.5$.*Why: Take the real cube root.***Final answer****h:** $h = 2.5$

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Write inverse-square proportionality as $A=k/r^2$.
2. Use $A=5$ when $r=.3$: $5=k/(.3)^2$, so $k=.45$.
3. Therefore $A=.45/r^2$.

Final answer: $A=0.45/r^2$

16 A is inversely proportional to the square of r

$$A = 5 \text{ when } r = 0.3$$

(a) Find a formula for A in terms of r

.....
(3)

(b) Find the value of A when $r = 7.5A$

$A =$
(3)

Worked solution

June 2021 | Question 16 | 3 marks

Family: Build an inverse-variation model • Variant: Inverse-square or inverse-power variation

Method / working

1. Substitute $r=7.5A$ into $A=.45/r^2$.
 2. Rearrange: $A^3=.45/(7.5^2)=.008$.
 3. Take the positive cube root: $A=.2$.
- Final answer: $A=0.2$

4MA1-2H Q20

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

WORKING SPACE

4MA1-2H Q20

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$
express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

Algebraic substitution

3 marks

Substitute

$$x = 1/(4y), \quad k = 1/(4y) - y.$$

Simplify

$$\frac{5k}{x+2} = \frac{5-20y^2}{1+8y}.$$

Final answer

$$\frac{5-20y^2}{1+8y}$$

4MA1-2H Q17

4 marks

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

.....
(4)

WORKING SPACE

4MA1-2H Q17

4 marks

(b) Make e the subject of $w = \sqrt{\frac{e+g}{ef-d}}$

.....
(4)

Make e the subject of a square-root formula

4 marks

Square

$$w^2 = \frac{e+g}{ef-d}.$$

Clear the denominator

$$w^2(ef-d) = e+g \Rightarrow w^2ef - w^2d = e+g.$$

Collect e

$$e(w^2f-1) = g+w^2d.$$

Divide

$$e = \frac{g+w^2d}{w^2f-1}.$$

Final answer

$$e = \frac{g+w^2d}{w^2f-1}$$

19 $ABCED$ is a five-sided shape.

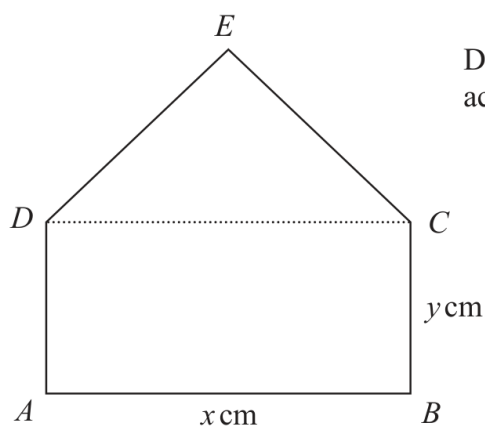


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

.....

.....

.....

(1)

Worked solution

November 2021 | Question 19 | 2 marks

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method / working

1. The perimeter is made from three sides of length (x) and two sides of length (y) :

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $3x+2y=100$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $y=\frac{100-3x}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The area is the area of the rectangle plus the area of the equilateral triangle:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $R=xy+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Substitute $(y=\frac{100-3x}{2})$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $R=x\left(\frac{100-3x}{2}\right)+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $R=50x-\frac{3}{2}x^2+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. For the maximum, write

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $R=-\frac{6-\sqrt{3}}{4}x^2+50x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. The maximum occurs at

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $x=\frac{-b}{2a}=\frac{-50}{2\left(-\frac{6-\sqrt{3}}{4}\right)}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $x=\frac{100}{6-\sqrt{3}}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. The coefficient of (x^2) is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. Answer: $(R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right))$, and $(x=\frac{100}{6-\sqrt{3}})$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$, and $x=\frac{100}{6-\sqrt{3}}$.

19 $ABCED$ is a five-sided shape.

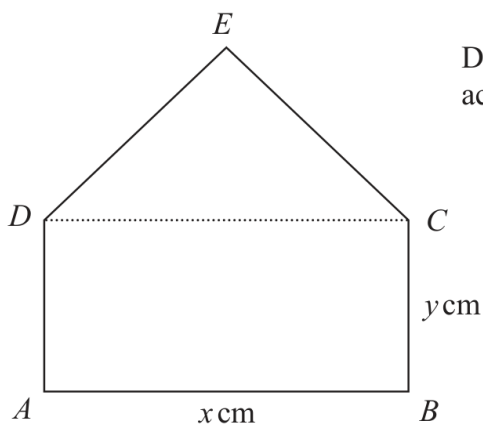


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} \left(200 - [6 - \sqrt{3}]x \right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

(1)

Worked solution

November 2021 | Question 19 | 1 marks

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method / working

1. The perimeter is made from three sides of length $\sqrt{x}$ and two sides of length $\sqrt{y}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $3x+2y=100$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $y=\frac{100-3x}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The area is the area of the rectangle plus the area of the equilateral triangle:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $R=xy+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Substitute $y=\frac{100-3x}{2}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $R=x\left(\frac{100-3x}{2}\right)+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $R=50x-\frac{3}{2}x^2+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. For the maximum, write

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $R=-\frac{6-\sqrt{3}}{4}x^2+50x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. The maximum occurs at

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $x=\frac{-b}{2a}=\frac{-50}{2\left(-\frac{6-\sqrt{3}}{4}\right)}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $x=\frac{100}{6-\sqrt{3}}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. The coefficient of (x^2) is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. Answer: $\left(R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)\right)$, and $\left(x=\frac{100}{6-\sqrt{3}}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$, and $x=\frac{100}{6-\sqrt{3}}$.

$$I = 8I^-$$

(Total for Question 16 is 4 marks)

WORKING SPACE

Make t the subject**4 marks****Clear the denominator**

$$c(7 - 8t^2) = t^2 + 3 \Rightarrow 7c - 8ct^2 = t^2 + 3.$$

Collect and factor

$$7c - 3 = (1 + 8c)t^2.$$

Take square roots

$$t^2 = \frac{7c - 3}{1 + 8c} \Rightarrow t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}.$$

Final answer

$$t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}$$

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

Worked solution

January 2020 | Question 16 | 4 marks

Family: Rearrange a subject inside one fraction • Variant: Subject in the denominator

Method / working

1. Square the equation: $y^2 = (x+1)/(x-4)$.
 2. Multiply through by $x-4$ to remove the fraction.
 3. Expand and collect x terms: $x(y^2-1) = 4y^2+1$.
 4. Divide by y^2-1 : $x = (4y^2+1)/(y^2-1)$.
- Final answer: $x = (4y^2+1)/(y^2-1)$

4MA1-2H Q18

3 marks

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

WORKING SPACE

4MA1-2H Q18

3 marks

- 18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

Chapter: Algebraic Proof

3 marks

Q18 – Chapter: Algebraic Proof

Family: Prove divisibility or a remainder • Variant: Fixed remainder modulo a divisor

MethodLet the consecutive odd numbers be $2n-1$ and $2n+1$.*Why: Any two consecutive odd numbers have this form.***Method**Their square sum is $(2n-1)^2 + (2n+1)^2 = 8n^2 + 2$.*Why: Expand and add.***Method** $8n^2$ is divisible by 8, so the remainder is always 2.*Why: Separate the multiple of 8 from the remainder.***Final answer**

remainder: 2

- 17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

Worked solution

January 2020 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Consecutive integer parity identity

Method / working

1. Let consecutive squares be n^2 and $(n+1)^2$.
2. Difference = $(n+1)^2 - n^2 = 2n+1$.
3. $2n+1$ is always odd for integer n .

Final answer: $(n+1)^2 - n^2 = 2n+1$, which is odd

- 17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

Worked solution

June 2021 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Consecutive even/odd sequence proof

Method / working

1. Represent consecutive even numbers as $2n$, $2n+2$, $2n+4$.
2. Subtract squares: $(2n+4)^2 - (2n)^2 = 16n+16$.
3. Factor: $16n+16 = 8(2n+2)$, which proves the result.

Final answer: Proof: difference = $8(2n+2)$

19 $ABCED$ is a five-sided shape.

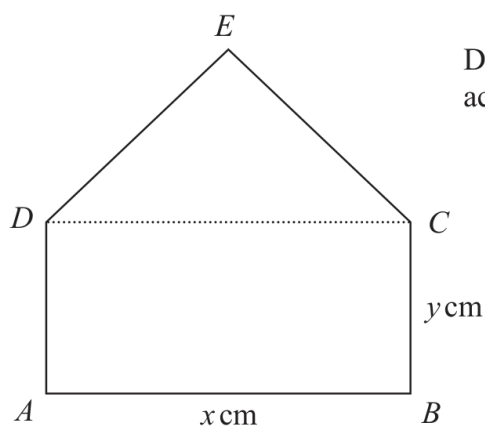


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

(a) Show that $R = \frac{x}{4} (200 - [6 - \sqrt{3}]x)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots$$

(2)

(ii) Explain why the maximum value of R is given by this value of x .

.....

.....

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(1)

Worked solution

November 2021 | Question 19 | 3 marks

Family: Construct an algebraic identity proof • Variant: Derive a general geometric algebra formula

Method / working

1. The perimeter is made from three sides of length $\sqrt{x}$ and two sides of length $\sqrt{y}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. $3x+2y=100$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. So

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. $y=\frac{100-3x}{2}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. The area is the area of the rectangle plus the area of the equilateral triangle:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. $R=xy+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. Substitute $y=\frac{100-3x}{2}$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $R=x\left(\frac{100-3x}{2}\right)+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. $R=50x-\frac{3}{2}x^2+\frac{\sqrt{3}}{4}x^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. For the maximum, write

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $R=-\frac{6-\sqrt{3}}{4}x^2+50x$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. The maximum occurs at

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. $x=\frac{-b}{2a}=\frac{-50}{2\left(-\frac{6-\sqrt{3}}{4}\right)}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $x=\frac{100}{6-\sqrt{3}}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. The coefficient of (x^2) is negative, so the quadratic graph opens downwards and this vertex gives a maximum.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. Answer: $\left(R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)\right)$, and $\left(x=\frac{100}{6-\sqrt{3}}\right)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $R=\frac{x}{4}\left(200-(6-\sqrt{3})x\right)$, and $x=\frac{100}{6-\sqrt{3}}$.

- 17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$
where k is an integer and m is a prime number,

(a) work out the value of k

$$k = \dots\dots\dots (1)$$

- (b) Show that $\frac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.

Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

Answer reference | January 2023 | Question 17

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Question	working	Answer	Mark	Notes
17 (a)		12	1	B1
(b)	$\frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{1+\sqrt{2}}{1+\sqrt{2}} \text{ or } \frac{5-\sqrt{18}}{1-\sqrt{2}} \times \frac{-1-\sqrt{2}}{-1-\sqrt{2}} \text{ oe}$		3	M1 Multiplying numerator and denominator by $1+\sqrt{2}$
	$\frac{5-\sqrt{36}+5\sqrt{2}-\sqrt{18}}{1+\sqrt{2}-\sqrt{2}-2} \text{ or } \frac{5-6-3\sqrt{2}+5\sqrt{2}}{-1} \text{ or}$ $\frac{-5+6+3\sqrt{2}-5\sqrt{2}}{1} \text{ oe}$ NB: allow $\sqrt{18}$ or $3\sqrt{2}$ $\sqrt{36}$ or 6 or $\sqrt{6}\sqrt{6}$			M1 Showing correct expansions (not necessarily as a fraction)
	working required	$1-2\sqrt{2}$		A1 dep on M2 (ie all stages of working must be shown convincingly) or for stating $a = 1$ and $b = -2$
Total 4 marks				

Worked solution

January 2023 | Question 17 | 4 marks

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Official final mark-scheme pages are embedded immediately after this question.

Show clear algebraic working.

.....
(Total for Question 19 is 3 marks)

WORKING SPACE

Quadratic inequality

3 marks

Factorise

$$4x^2 + 4x - 15 = (2x - 3)(2x + 5).$$

Use the sign of an upward quadratic

The expression is negative between the roots $-\frac{5}{2}$ and $\frac{3}{2}$.

Final answer

$$-2.5 < x < 1.5$$

4MA1-2H Q22

5 marks

22 Here is a rectangle.

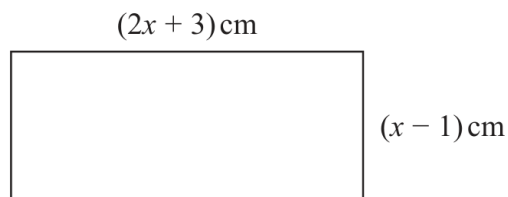


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

WORKING SPACE

4MA1-2H Q22

5 marks

22 Here is a rectangle.

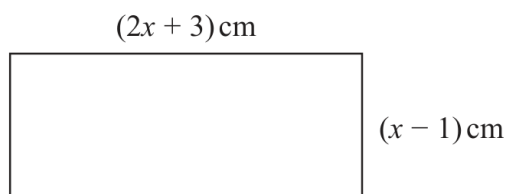


Diagram **NOT**
accurately drawn

Given that the area of the rectangle is less than 75 cm^2

find the range of possible values of x

Chapter: Solving Inequalities

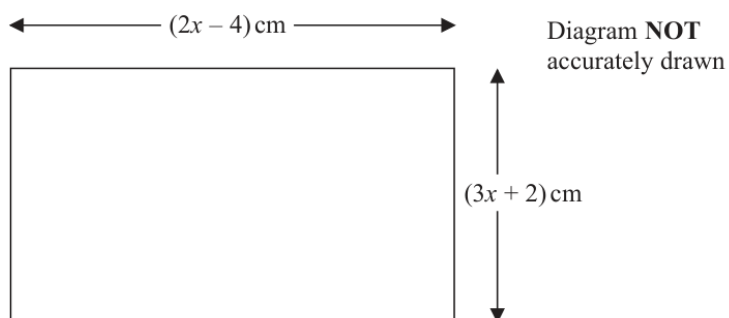
5 marks

Q22 – Chapter: Solving Inequalities

Family: Solve quadratic inequalities • Variant: Contextual quadratic feasible range

Method $(2x+3)(x-1) < 75$ gives $2x^2 + x - 78 < 0$.*Why: Expand and move 75 to the left.***Method**The critical values solve $2x^2 + x - 78 = 0 = (x-6)(2x+13)$, so $x=6$ or $x=-6.5$.*Why: Factorise the boundary equation.***Method**Because the rectangle side $x-1$ must be positive, $x > 1$.*Why: Lengths are positive.***Method**The quadratic is negative between its roots, so $x < 6$.*Why: Use the sign of the upward-opening quadratic.***Method**Therefore $1 < x < 6$.*Why: Combine the restrictions.***Final answer**range: $1 < x < 6$

17 The diagram shows a rectangle.



The area of the rectangle is $A \text{ cm}^2$

Given that $A < 3x + 27$

find the range of possible values for x .

(Total for Question 17 is 5 marks)

Answer reference | May-Nov 2020 | Question 17

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

17	$(3x+2)(2x-4) < 3x+27$ or eg $6x^2 - 8x - 8 < 3x + 27$ eg $6x^2 - 11x - 35 < 0$			M1	condone incorrect symbol
				M1	expanding and rearranging to get a correct 3 term quadratic, condone incorrect symbol
	$(2x-7)(3x+5) (= 0)$ or $\frac{11 \pm \sqrt{(-11)^2 - 4 \times 6 \times (-35)}}{2 \times 6}$			M1	first step to find the critical values dep on M1 for solving their 3 term quadratic using any correct method (allow one sign error and some simplification – allow as far as the equivalent of $\frac{11 \pm \sqrt{121+840}}{12}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$-\frac{5}{3}, \frac{7}{2}$			A1	oe the positive critical value only or both critical values (if both they must be correct)
		$2 < x < \frac{7}{2}$	5	A1	accept $2 \leq x < \frac{7}{2}$ may be seen as two separate inequalities $x > 2$ ($x \leq 2$) and $x < \frac{7}{2}$
				Total 5 marks	

Worked solution

May-Nov 2020 | Question 17 | 5 marks

Family: Translate inequality notation and number lines • Variant: Inequality symbol from a number line

Official final mark-scheme pages are embedded immediately after this question.

- 22** Solve the inequality $6x^2 + 37x \leq 35$
Show clear algebraic working.

.....
(Total for Question 22 is 3 marks)

Answer reference | January 2023 | Question 22

Family: Solve linear inequalities • Variant: One-sided linear inequality

Question	working	Answer	Mark	Notes
22	$(6x - 5)(x + 7)(=0) \text{ or }$ $\frac{-37 \pm \sqrt{37^2 - 4 \times 6 \times -35}}{2 \times 6}$ $6 \left[\left(x + \frac{37}{12} \right)^2 - \left(\frac{37}{12} \right)^2 \right] \dots \text{ oe}$		3	M1 A correct method to solve the quadratic equation $6x^2 + 37x - 35(=0)$ using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error in substitution and some simplification – allow as far as $\frac{-37 \pm \sqrt{1369 + 840}}{12}$) or completing the square as far as shown on left
	$\frac{5}{6}$ oe and -7			A1 dep on M1 correct critical values (allow 0.83...)
	Working must be seen for both accuracy marks as asked for in question	$-7 \leq x \leq \frac{5}{6}$		A1 dep on M1 oe eg $-7 \leq x \leq 0.83\dots$, $\left[-7, \frac{5}{6} \right]$ Accept $x \leq \frac{5}{6}, x \geq -7$
Total 3 marks				

Worked solution

January 2023 | Question 22 | 3 marks

Family: Solve linear inequalities • Variant: One-sided linear inequality

Official final mark-scheme pages are embedded immediately after this question.

4MA1-2H Q22

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

WORKING SPACE

4MA1-2H Q22

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

Line-curve intersections

5 marks

Substitute

$$y = 5 - x : \quad 2x^2 + 3(5 - x)^2 = 210.$$

Solve

$$x^2 - 6x - 27 = 0 \Rightarrow x = 9, -3.$$

Coordinates

$$y = -4, 8.$$

Final answer

$$(9, -4), (-3, 8)$$

4MA1-2H Q22 M01

5 marks

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

WORKING SPACE

Solve simultaneous equations

5 marks

M01 – Solve simultaneous equations

Substitute $y=x+2$

$$x^2 + y^2 + y = 3 \Rightarrow x^2 + (x+2)^2 + (x+2) = 3.$$

Form the quadratic

$$2x^2 + 5x + 3 = 0 = (2x+3)(x+1).$$

Find x-values

$$x = -3/2 \text{ or } x = -1.$$

Find y-values

$$y = x + 2 \Rightarrow y = 1/2 \text{ or } y = 1.$$

Pair the solutions

$$(-3/2, 1/2) \text{ and } (-1, 1).$$

Final answer

$$\left(-\frac{3}{2}, \frac{1}{2}\right) \text{ and } (-1, 1)$$

- 22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .

Show clear algebraic working.

(..... ,)

(..... ,)

Worked solution

January 2020 | Question 22 | 5 marks

Family: Solve a linear-quadratic system • Variant: Multiple solution-pair validation

Method / working

1. Substitute $y=x+2$ into $x^2+y^2-2y=24$.
2. Obtain $x^2+x-12=0$.
3. Factor: $(x+4)(x-3)=0$.
4. $x=-4$ gives $y=-2$; $x=3$ gives $y=5$.
5. State both points.

Final answer: $(-4,-2)$ and $(3,5)$

16 Solve the simultaneous equations

$$\begin{aligned} 3xy - y^2 &= 8 \\ x - 2y &= 1 \end{aligned}$$

Show clear algebraic working.

(Total for Question 16 is 5 marks)

Answer reference | May-Nov 2020 | Question 16

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

16	$3y(2y+1) - y^2 = 8$ or $x = \frac{8+y^2}{3y} \rightarrow \frac{8+y^2}{3y} - 2y = 1$ or $-3xy - y^2 = 8$ $3xy - 3y \times 2y = 3y \times 1$ oe	$3x\left(\frac{x-1}{2}\right) - \left(\frac{x-1}{2}\right)^2 = 8$ oe			M1 correct first step eg substitution by eg $x = 1 + 2y$ or $y = \frac{x-1}{2}$ to get an equation in a single variable or writing 2 nd equation with x the subject and substituting into 1 st or multiplying 2 nd equation by $3y$ and subtracting from 1 st oe
	eg $5y^2 + 3y - 8 (= 0)$	eg $5x^2 - 4x - 33 (= 0)$			A1 for a correct simplified quadratic
	$(5y+8)(y-1) (= 0)$ or $\frac{-3 \pm \sqrt{3^2 - 4 \times 5 \times (-8)}}{2 \times 5}$	$(5x+11)(x-3) (= 0)$ or $\frac{4 \pm \sqrt{(-4)^2 - 4 \times 5 \times (-33)}}{2 \times 5}$			M1ft dep on M1 for solving their 3 term quadratic equation using any correct method (allow one sign error and some simplification – allow as far as $\frac{-3 \pm \sqrt{9+160}}{10}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$y = -\frac{8}{5}$ and $y = 1$ (both)	$x = -\frac{11}{5}$ and $x = 3$ (both)			A1 dep on first M1
		$x = -\frac{11}{5}, y = -\frac{8}{5}$ $x = 3, y = 1$	5		A1 oe dep on first M1 Must be paired correctly
					Total 5 marks

Worked solution

May-Nov 2020 | Question 16 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

20 Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

20	$x^2 + (7 - 2x)^2 = 34$	$\left(\frac{7-y}{2}\right)^2 + y^2 = 34$	5	M1	substitution of linear equation into quadratic
	$5x^2 - 28x + 15 [= 0]$ oe	$5y^2 - 14y - 87 [= 0]$ oe		M1	dep on previous M1 for multiplying out and collecting terms, forming a three term quadratic in any form of $ax^2 + bx + c (= 0)$ where at least 2 coefficients (a or b or c) are correct and all are non-zero
	$(5x - 3)(x - 5) [= 0]$ or $\frac{-(-28) \pm \sqrt{(-28)^2 - 4 \times 5 \times 15}}{2 \times 5}$ or $5\left(x - \frac{28}{10}\right)^2 - \frac{784}{100} + 15 = 0$ oe or $x = 0.6$ and $x = 5$ (allow incorrect labels for x/y)	$(5y - 29)(y + 3) [= 0]$ or $\frac{-(-14) \pm \sqrt{(-14)^2 - 4 \times 5 \times (-87)}}{2 \times 5}$ or $5\left(y - \frac{14}{10}\right)^2 - \frac{196}{100} - 87 = 0$ oe or $y = 5.8$ and $y = -3$ (allow incorrect labels for x/y)		M1ft	dep on M1 for solving <i>their</i> 3 term quadratic equation using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error and some simplification – allow as far as $\frac{28 \pm \sqrt{784 - 300}}{10}$ or $\frac{14 \pm \sqrt{196 + 1740}}{10}$) (if completing the square allow as far as shown) or correct values for x or correct values for y dep on correct quadratic
	eg $y = 7 - 2 \times 5$ and $y = 7 - 2 \times 0.6$ (correct labels for x/y)	eg $5.8 = 7 - 2x$ and $-3 = 7 - 2x$ (correct labels for x/y)		M1ft	dep on previous M1 for substituting their 2 found values of x or y in a suitable equation or correct values for the other variable
	<i>Working must be shown</i>		$x = 0.6, y = 5.8$ $x = 5, y = -3$	A1	dep on M1 and the correct quadratic (allow coordinates) must be paired correctly

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

				Total 5 marks
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Worked solution

January 2023 | Question 20 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q24

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

WORKING SPACE

4MA1-1H Q24

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

Arithmetic sequence of polygon angles

6 marks

Two sums

$$S_n = \frac{n}{2}[168 + 4(n - 1)] = 2n^2 + 82n, \quad S_n = 180(n - 2).$$

Solve

$$n^2 - 49n + 180 = 0 \Rightarrow n = 45 \ (n > 5).$$

Angle sum

$$180(45 - 2) = 7740.$$

Final answer

$$7740^\circ$$

4MA1-1H Q19 demand

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q19 demand

6 marks

19 Here are the first 4 terms in an arithmetic sequence.

3 7 11 15

The last term of the sequence is x

The sum of the terms of the sequence is 7260

Find the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

Last term of an arithmetic sequence

6 marks

Demand – Last term of an arithmetic sequence

Identify a and d

$$a = 3, \quad d = 4.$$

Use the sum

$$S_n = \frac{n}{2}[2(3) + (n-1)4] = 7260 \Rightarrow 2n^2 + n - 7260 = 0.$$

Solve for the number of terms

$$(2n + 121)(n - 60) = 0 \Rightarrow n = 60.$$

Find the last term

$$x = 3 + (60 - 1)4 = 239.$$

Final answer

239

4MA1-1H Q19 M01

5 marks

19 An arithmetic series has first term a and common difference d

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.
Show clear algebraic working.

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WORKING SPACE

Find the sum of an arithmetic series

5 marks

M01 – Find the sum of an arithmetic series**Use the sum of 30 terms**

$$30a + 435d = 4395 \Rightarrow 2a + 29d = 293.$$

Use the 10th and 20th terms

$$(a + 9d) + (a + 19d) = 284 \Rightarrow 2a + 28d = 284.$$

Eliminate

$$d = 9, \quad a = 16.$$

Use the sum formula

$$S_{45} = \frac{45}{2}[2(16) + 44(9)] = 9630.$$

Final answer

9630

24 An arithmetic series has first term a and common difference d .

The sum of the first $2n$ terms of the series is four times the sum of the first n terms of the series.

Find an expression for a in terms of d .

Show your working clearly.

$a = \dots\dots\dots$

Worked solution

January 2021 | Question 24 | 4 marks

Family: Recover arithmetic-series parameters • Variant: Symbolic relation between partial sums

Method / working

1. Write $S_n = n[2a + (2n-1)d]$.
2. Set S_n equal to $4S_n$ and cancel n .
3. Simplify to $2a - d = 4a - 2d$.
4. Therefore $a = d/2$.

Final answer: $a = d/2$

24 Here are the first five terms of an arithmetic sequence.

8 15 22 29 36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

Answer reference | May-Nov 2020 | Question 24

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

24	$a = 8 \quad d = 7$		4	M1 can be implied
	$(S_{100} =) \frac{100}{2} (2 \times 8 + (100 - 1) \times 7) (= 35\,450) \text{ or}$ $(S_{49} =) \frac{49}{2} (2 \times 8 + (49 - 1) \times 7) (= 8624) \text{ or}$ $(S_{50} =) \frac{50}{2} (2 \times 8 + (50 - 1) \times 7) (= 8975)$			M1
	'35450' – '8624' or '35450' – '8975' + $(8 + (50 - 1) \times 7)$			M1
		26 826		A1
				Total 4 marks
	Alternative scheme			
	$(u_n =) 7n + 1$	$a = 8 \text{ and } d = 7$	4	M1 can be implied
	$(u_{50} =) 7 \times 50 + 1 (= 351) \text{ or}$ $(u_{100} =) 7 \times 100 + 1 (= 701)$	$(u_{50} =) 8 + (50 - 1) \times 7$ $(= 351)$		M1
	$\frac{51}{2} ('351' + '701')$	$\frac{51}{2} (2 \times 351 + (51 - 1) \times 7)$		M1
		26 826		A1
				Total 4 marks

Worked solution

May-Nov 2020 | Question 24 | 4 marks

Family: Use an arithmetic nth term • Variant: Arithmetic nth term from listed terms

Method

The arithmetic sequence is

8, 15, 22, 29, 36, ...

So

$a=8$, $d=7$

The nth term is

$u_n = 8 + 7(n-1) = 7n + 1$

$u_{50} = 7(50) + 1 = 351$

$u_{100} = 7(100) + 1 = 701$

There are

$100 - 50 + 1 = 51$

terms from the 50th to the 100th inclusive.

$\text{sum} = \frac{51}{2}(351 + 701)$

$= 26826$

Answer: 26826

$$g(x) = 4x^2$$

(a) Find $f(9)$

(1)

WORKING SPACE

Function substitution

1 marks

Substitute

$$f(9) = 9 - \sqrt{9} = 9 - 3 = 6.$$

Final answer

$$f(9) = 6$$

$$g(x) = 4x^2$$

- (b) Solve $fg(x) < 0$
Show clear algebraic working.

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(3)

WORKING SPACE

Composite function inequality

3 marks

Compose the functions rigorously

$$fg(x) = f(4x^2) = 9 - \sqrt{4x^2} = 9 - 2|x|.$$

Solve the inequality as written

$$9 - 2|x| < 0 \Rightarrow |x| > \frac{9}{2} \Rightarrow x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

Read the domain statement carefully

The printed $x \geq 0$ is the input domain of f . Since $g(x) = 4x^2 \geq 0$ for every real x , it does not remove the negative branch.

If the original variable is additionally restricted

If the original variable is also restricted to $x \geq 0$, then $|x| = x$, so the result reduces to $x > \frac{9}{2}$.

Final answer

$$\text{As written: } x < -\frac{9}{2} \text{ or } x > \frac{9}{2}.$$

$$\text{With an additional } x \geq 0 \text{ restriction: } x > \frac{9}{2}.$$

4MA1-1H Q17

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

WORKING SPACE

4MA1-1H Q17

4 marks

17 The functions g and h are such that

$$g(x) = \frac{11}{2x - 5}$$

$$h(x) = x^2 + 4 \quad x \geq 0$$

(a) What value of x must be excluded from any domain of g ?

.....
(1)

(b) Solve $gh(x) = 1$

.....
(3)

Domain exclusion and composite equation

4 marks

Exclude a zero denominator

$$2x - 5 \neq 0 \Rightarrow x \neq \frac{5}{2}.$$

Form the composite

$$g(h(x)) = \frac{11}{2(x^2 + 4) - 5} = 1.$$

Solve with the stated nonnegative domain

$$2x^2 + 3 = 11 \Rightarrow x^2 = 4, \quad x = 2.$$

Final answer

$$x \neq \frac{5}{2}; \quad x = 2 \text{ for } g(h(x)) = 1$$

4MA1-2H Q25 Q25(b)

1 marks

(b) State the domain of g^{-1}

(1)

WORKING SPACE

4MA1-2H Q25 Q25(b)

1 marks

(b) State the domain of g^{-1}

.....
(1)

Chapter: Functions

1 marks

Q25(b) – Chapter: Functions

Family: Control domain and range • Variant: Domain and range swap under inversion

Method

The square root requires $14-x \geq 0$, so the domain of g^{-1} is $x \leq 14$.

Why: Use the radicand condition.

Final answer

domain: $x \leq 14$

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

.....
(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 2 marks

Family: Compose functions in the stated order • Variant: Evaluate a numerical composite function

Method / working

1. $g(2) = 7 \times 2 - 6 = 8.$

2. $f(g(2)) = f(8) = 5(8)^2 - 10(8) + 7 = 247.$

Final answer: 247

24 The functions f and g are defined as

$$f(x) = 5x^2 - 10x + 7 \quad \text{where } x \geq 1$$

$$g(x) = 7x - 6$$

(a) Find $fg(2)$

.....
(2)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(4)

Worked solution

June 2021 | Question 24 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a restricted quadratic function

Method / working

1. Write $y = 5x^2 - 10x + 7 = 5(x-1)^2 + 2$ with $x \geq 1$.
 2. Rearrange: $(x-1)^2 = (y-2)/5$.
 3. Use the domain $x \geq 1$, so $x-1$ is the positive square root.
 4. Swap x and y : $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$.
- Final answer: $f^{-1}(x) = 1 + \sqrt{(x-2)/5}$

18 The function f is such that $f(x) = \frac{k}{x}$ where $x \neq 0$ and k is an integer.

(a) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$$f^{-1}(x) = \dots \quad (1)$$

The function g is such that $g(x) = 2 - 3x^4$ where $x \neq 0$

The function h is such that $h(x) = \frac{3x}{2-x}$ where $x \neq 2$

(b) (i) Find $g(-2)$

$$\dots \quad (1)$$

(ii) Express the composite function hg in the form $hg(x) = \dots$
Give your answer in its simplest form.

$$hg(x) = \dots \quad (2)$$

(Total for Question 18 is 4 marks)

Answer reference | January 2023 | Question 18

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

18	(a)		$\frac{k}{x}$	1	B1 allow kx^{-1}
	(b)(i)		-46	1	B1 cao
	(ii)	$\frac{3(2-3x^4)}{2-(2-3x^4)}$ or $\frac{6-9x^4}{2-2+3x^4}$ oe or $\frac{6-9x^4}{3x^4}$ oe		2	M1
		<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	$\frac{2-3x^4}{x^4}$		A1 allow $\frac{2}{x^4} - 3$ oe
					Total 4 marks

Worked solution

January 2023 | Question 18 | 4 marks

Family: Construct an inverse function • Variant: Inverse of a composite-defined function

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q23 M01

5 marks

- 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.
- C** and **L** intersect at the points A and B
- The coordinates of point A are (p, k)
The coordinates of point B are (q, k)
- Given that $p - q = 14$
- find the value of k
Show clear algebraic working.

4MA1-1H Q23 M01

5 marks

$k = \dots\dots\dots$

WORKING SPACE

Use the chord separation to find k**5 marks****M01 – Use the chord separation to find k****Use the equal y-values**

$$p^2 - 8p - 9 = k \text{ and } q^2 - 8q - 9 = k.$$

Use symmetry

The axis is $x=4$, so $(p+q)/2=4$; with $p-q=14$, $p=11$, $q=-3$.

Substitute

$$k = 11^2 - 8(11) - 9 = 24.$$

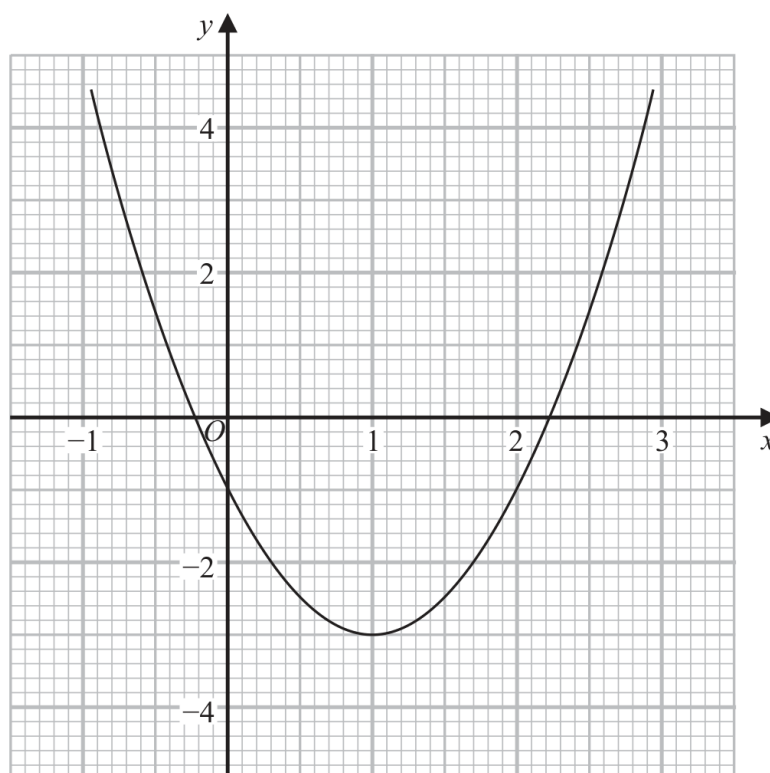
Final answer

$$k = 24$$

4MA1-2H Q21 Q21(a)

2 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....
(2)

Continue your method

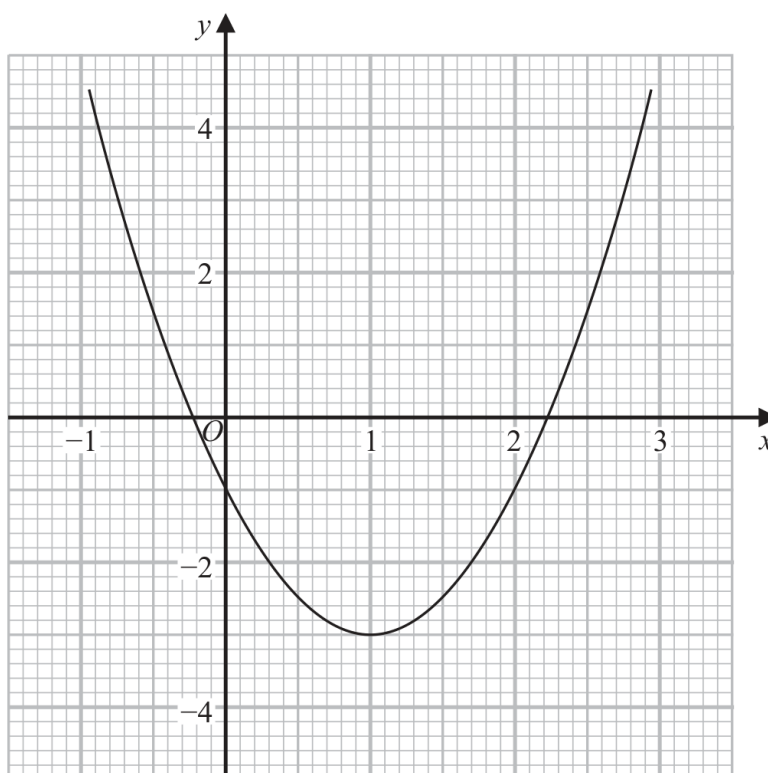
2 marks

WORKING SPACE

4MA1-2H Q21 Q21(a)

2 marks

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$.
Give your solutions correct to one decimal place.

.....
(2)

Chapter: Graphs of Functions

2 marks

Q21(a) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Estimate roots from axis crossings

Method

The x-intercepts of $y=2x^2-4x-1$ are approximately $x=-0.2$ and $x=2.2$.

Why: Read where the curve crosses the x-axis.

Method

Both values are given to one decimal place.

Why: Round the graph readings as requested.

Final answer

roots: -0.2 and 2.2

4MA1-2H Q21 Q21(b)

3 marks

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

4MA1-2H Q21 Q21(b)

3 marks

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$
Show your working clearly.
Give your solutions correct to one decimal place.

.....
(3)

Chapter: Graphs of Functions

3 marks

Q21(b) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

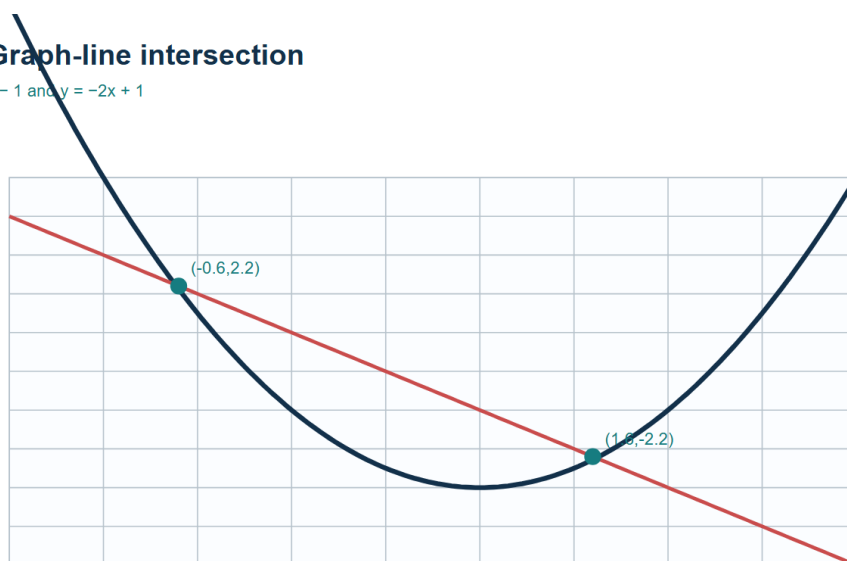
MethodFor $y=x^2-x-1$, use $y=-2x+1$ as the straight line.*Why: Rearrange $x^2-x-1=0$ to $x^2-x=1$ and compare with the displayed quadratic.***Method**Draw $y=-2x+1$ through $(0,1)$ and $(-1,3)$.*Why: Use two points to draw the line on the supplied grid.***Method**The intersections give $x \approx -0.6$ and $x \approx 1.6$.*Why: Read the two x -coordinates to one decimal place.***Final answer**roots: -0.6 and 1.6

Chapter: Graphs of Functions

3 marks

Q21 • Graph-line intersection

$$y = 2x^2 - 4x - 1 \text{ and } y = -2x + 1$$

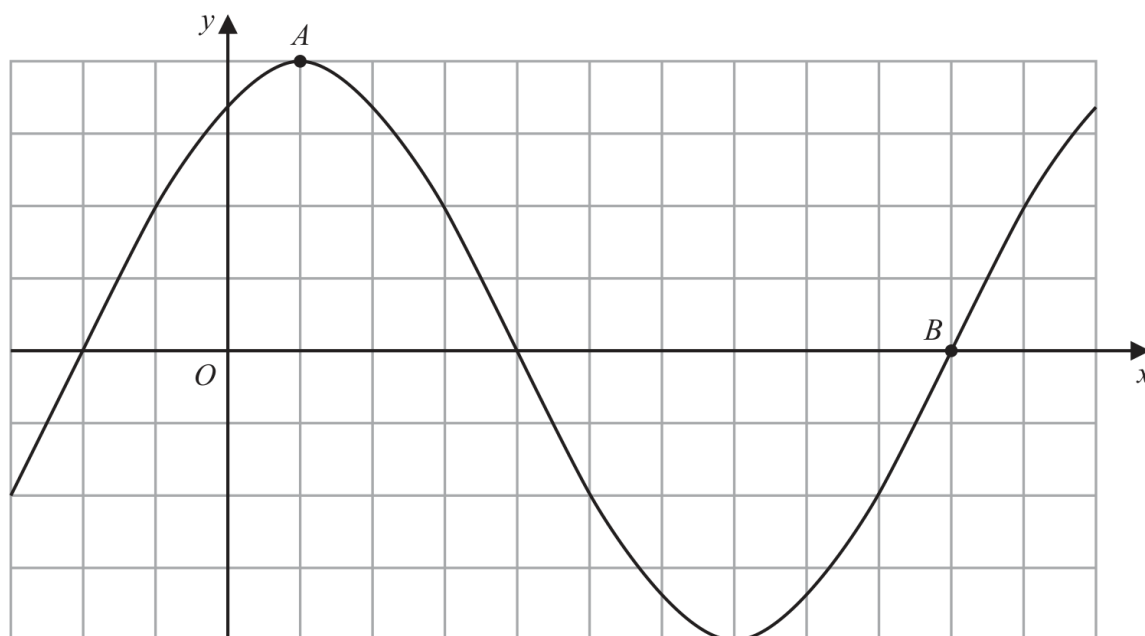
x-intersections: -0.6 and 1.6

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4MA1-2H Q25

2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

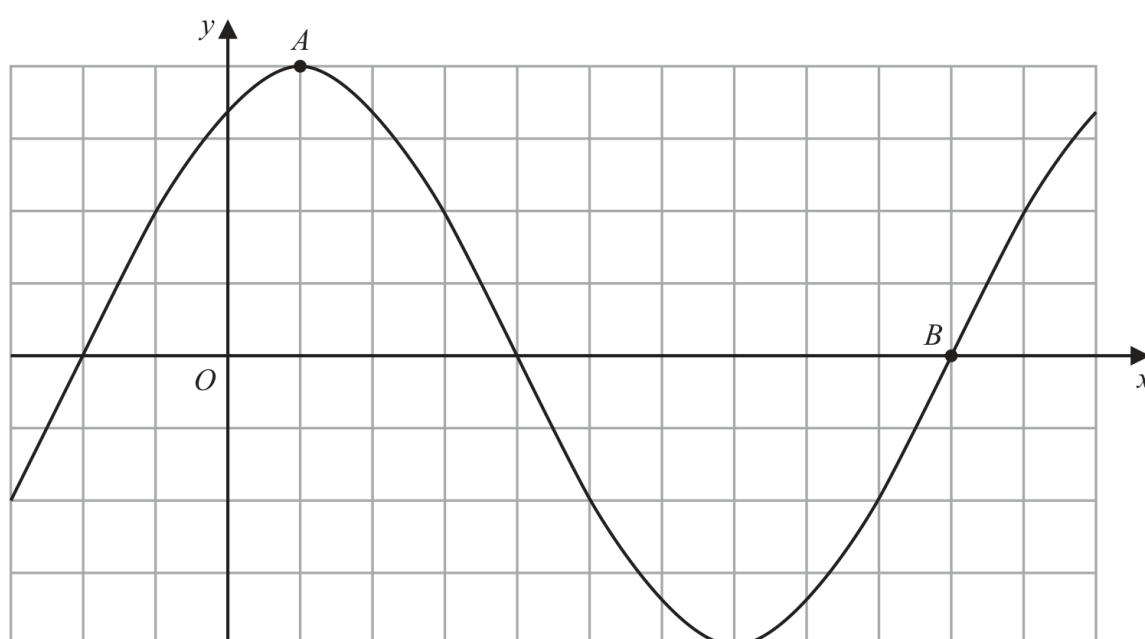
Continue your method**2 marks**

WORKING SPACE

4MA1-2H Q25

2 marks

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

Transformed sine graph

2 marks

Maximum

$$x + 60 = 90 \Rightarrow A = (30, 2).$$

Axis crossing

$$x + 60 = 360 \Rightarrow B = (300, 0).$$

Final answer

$$A = (30, 2), \quad B = (300, 0)$$

4MA1-2H Q19 M02

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.
The coordinates of this minimum point are (5, 4)

(ii) $y = 3f(x)$

(.....,)
(1)

WORKING SPACE

Stretch graph values**1 marks****M02 – Stretch graph values****Multiply the output**

For $y=3f(x)$, the x-coordinate stays 5 and the y-coordinate becomes $3(4)=12$.

Final answer

(5,12)

4MA1-2H Q19 M03

1 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

(iii) $y = f(x) - 7$

(.....,)
(1)

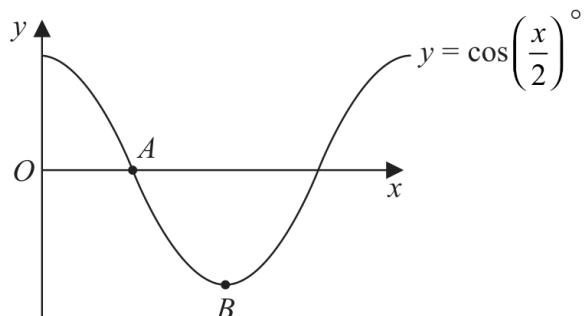
WORKING SPACE

Shift a graph vertically**1 marks****M03 – Shift a graph vertically****Subtract 7 from y**For $y=f(x)-7$, $(5,4)$ becomes $(5,4-7)=(5,-3)$.**Final answer** $(5,-3)$

4MA1-2H Q23 Q23(ii)

1 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(ii) Find the coordinates of the point B

(.....,)
(1)

Continue your method

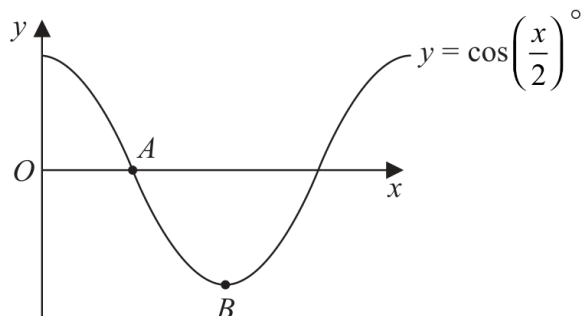
1 marks

WORKING SPACE

4MA1-2H Q23 Q23(ii)

1 marks

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(ii) Find the coordinates of the point B

(.....,)
(1)

Chapter: Transformations of Graphs

1 marks

Q23(ii) – Chapter: Transformations of Graphs

Family: Map a known graph feature • Variant: Horizontal scale or y-axis reflection

Method

At the first minimum, $x/2 = 180$ degrees, so $x = 360$ degrees and $y = -1$; hence $B = (360, -1)$.

Why: Use the first minimum of the cosine graph in degree mode.

Final answer

B: $B = (360, -1)$

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

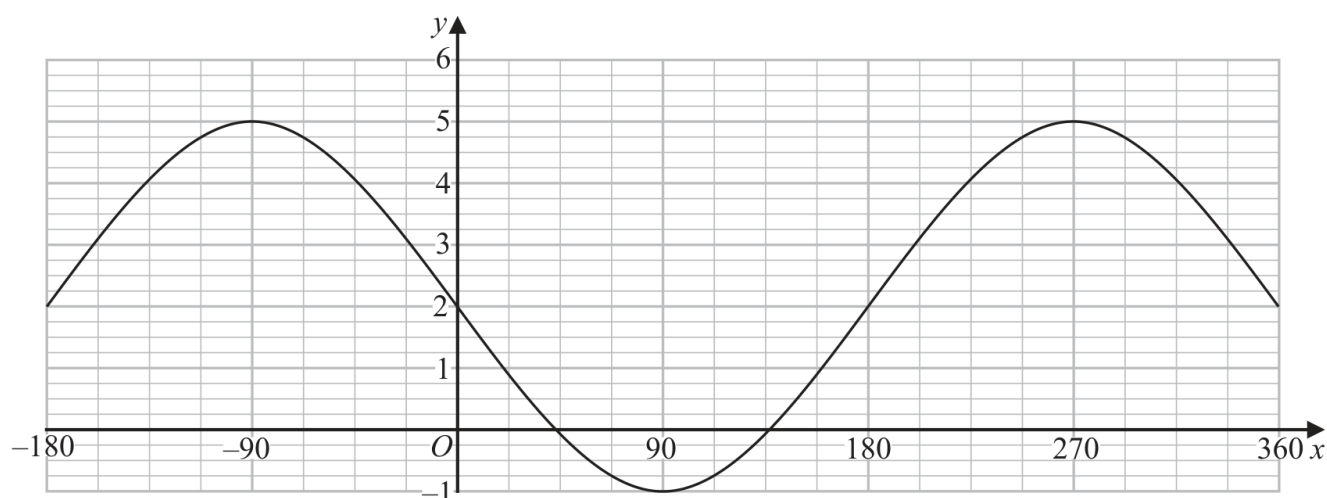
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation • Variant: Transform an explicit polynomial and simplify

Method / working

1. $f(x)=9-3(x+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. This is in completed square form, so the maximum point is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $(-2,9)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. For the translation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. the graph moves 4 units right. Replace $\mathcal{X}$ by $\mathcal{X}-4$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $y=9-3((x-4)+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $y=9-3(x-2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. The curve $\mathcal{T}$ has equation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $y=3(x+2)^2-9$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. This is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $y=-\left(9-3(x+2)^2\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. So $\mathcal{C}$ is reflected in the $\mathcal{X}$ -axis.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. For the cosine graph, the maximum is 5 and the minimum is (-1) .

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $a=\frac{5-(-1)}{2}=3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. $c=\frac{5+(-1)}{2}=2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. For

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. $y=a\cos(x-b)^{\circ}+c$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. a maximum occurs when $\mathcal{X}-b=0$. The graph has a maximum at $\mathcal{X}=270^{\circ}$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. $b=270$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. Answer: $\mathcal{A}=(-2,9)$, $\mathcal{Y}=9-3(x-2)^2$, reflection in the $\mathcal{X}$ -axis, $\mathcal{A}=3$, $b=270$, $c=2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $\mathcal{A}=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3$, $b=270$, $c=2$

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

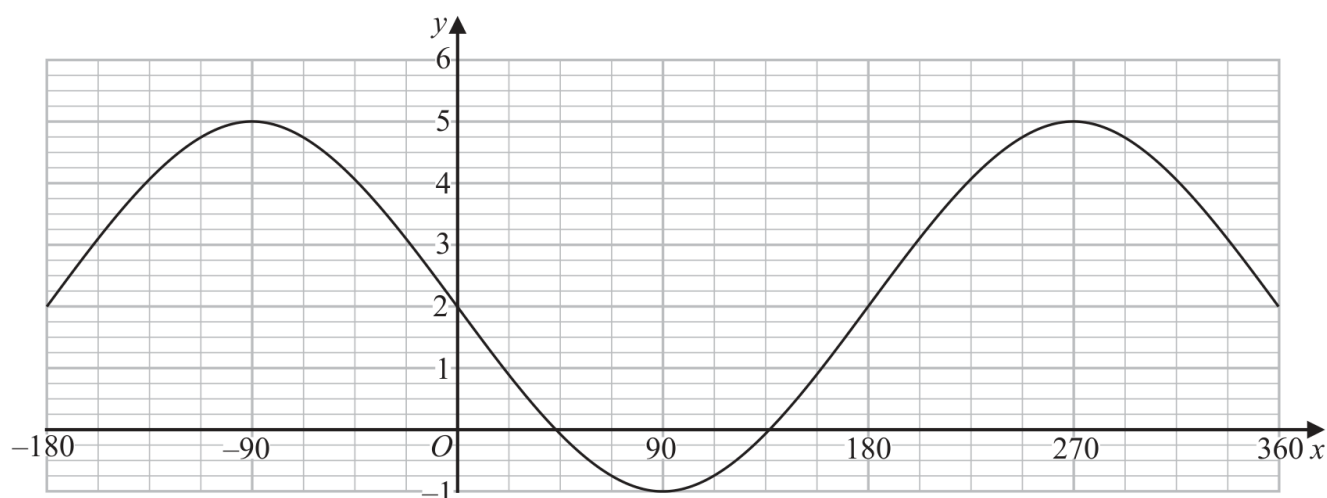
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(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Write the transformed equation • Variant: Translation vector applied to $y=f(x)$

Method / working

1. $f(x)=9-3(x+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. This is in completed square form, so the maximum point is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $(-2,9)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. For the translation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. the graph moves 4 units right. Replace (x) by $(x-4)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $y=9-3((x-4)+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $y=9-3(x-2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. The curve (T) has equation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $y=3(x+2)^2-9$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. This is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $y=-\left(9-3(x+2)^2\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. So (C) is reflected in the (x) -axis.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. For the cosine graph, the maximum is 5 and the minimum is (-1) .

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

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Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. For

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. $y=a\cos(x-b)^\circ+c$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. a maximum occurs when $(x-b=0)$. The graph has a maximum at $(x=270^\circ)$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. $b=270$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. Answer: $(A=(-2,9))$, $(y=9-3(x-2)^2)$, reflection in the (x) -axis, $(a=3, b=270, c=2)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x -axis, $a=3, b=270, c=2$

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

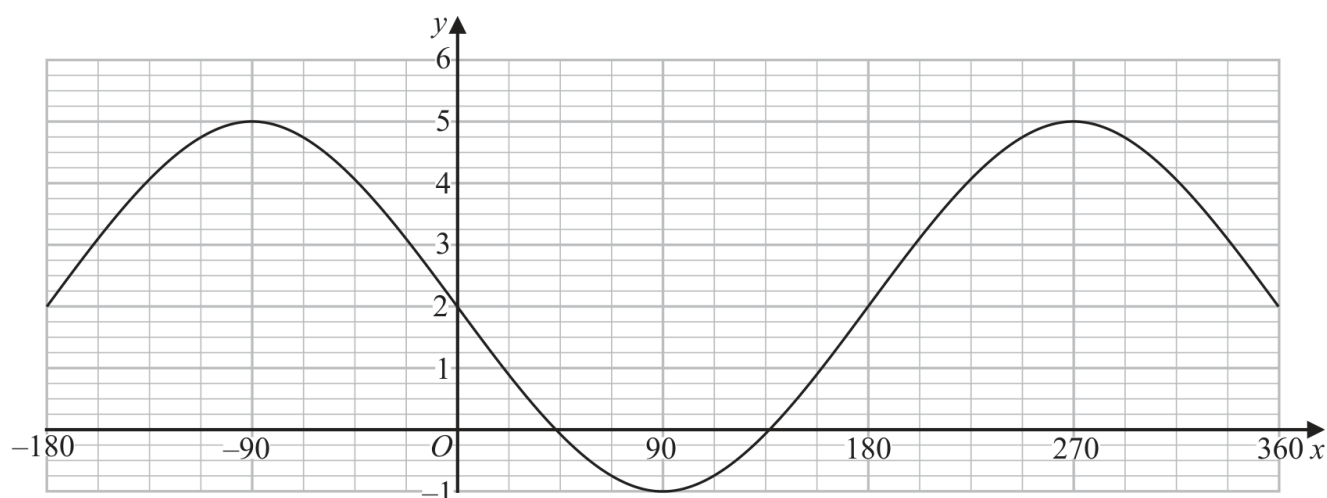
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(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 1 marks

Family: Infer or describe the graph transformation • Variant: Describe from two curve equations

Method / working

1. $f(x) = 9 - 3(x+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. This is in completed square form, so the maximum point is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $(-2, 9)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. For the translation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. the graph moves 4 units right. Replace x by $(x-4)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $y = 9 - 3((x-4)+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $y = 9 - 3(x-2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. The curve (T) has equation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $y = 3(x+2)^2 - 9$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. This is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $y = -\left(9 - 3(x+2)^2\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. So (C) is reflected in the x -axis.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. For the cosine graph, the maximum is 5 and the minimum is (-1) .

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $a = \frac{5 - (-1)}{2} = 3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. $c = \frac{5 + (-1)}{2} = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. For

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. $y = a \cos(x-b)^\circ + c$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. a maximum occurs when $(x-b=0)$. The graph has a maximum at $(x=270^\circ)$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. $b = 270$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. Answer: $A = (-2, 9)$, $y = 9 - 3(x-2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $A = (-2, 9)$, $y = 9 - 3(x-2)^2$, reflection in the x -axis, $a = 3$, $b = 270$, $c = 2$

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The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

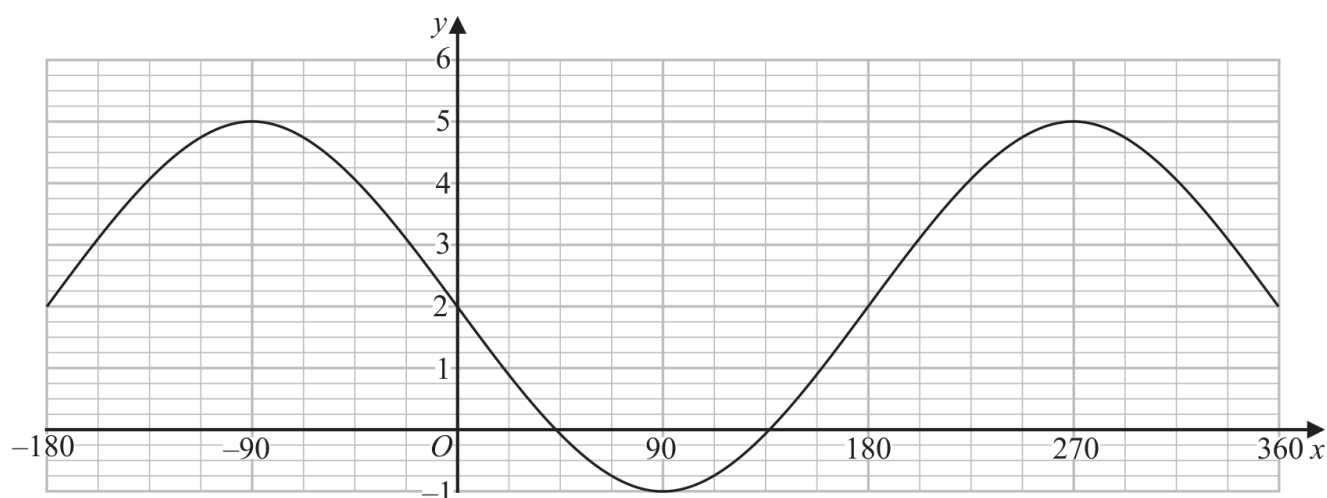
.....
(1)

The curve **C** is transformed to the curve **T**.
The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....
(1)

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$
 $b =$
 $c =$
(3)

Worked solution

November 2021 | Question 21 | 3 marks

Family: Map a known graph feature • Variant: Combined feature mapping

Method / working

1. $f(x)=9-3(x+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

2. This is in completed square form, so the maximum point is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

3. $(-2,9)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

4. For the translation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

5. $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

6. the graph moves 4 units right. Replace $\backslash(x)$ by $\backslash(x-4)$:

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

7. $y=9-3((x-4)+2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

8. $y=9-3(x-2)^2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

9. The curve $\backslash(T)$ has equation

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

10. $y=3(x+2)^2-9$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

11. This is

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

12. $y=-\backslashleft(9-3(x+2)^2\right)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

13. So $\backslash(C)$ is reflected in the $\backslash(x)$ -axis.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

14. For the cosine graph, the maximum is 5 and the minimum is $\backslash(-1)$.

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

15. $a=\frac{5-(-1)}{2}=3$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

16. $c=\frac{5+(-1)}{2}=2$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

17. For

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

18. $y=a\cos(x-b)^{\circ}+c$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

19. a maximum occurs when $\backslash(x-b=0)$. The graph has a maximum at $\backslash(x=270^{\circ})$, so

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

20. $b=270$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

21. Answer: $\backslash(A=(-2,9))$, $\backslash(y=9-3(x-2)^2)$, reflection in the $\backslash(x)$ -axis, $\backslash(a=3, \backslash b=270, \backslash c=2)$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $A=(-2,9)$, $y=9-3(x-2)^2$, reflection in the x-axis, $a=3, \backslash b=270, \backslash c=2$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

.....
(4)

(Total for Question 17 is 6 marks)

WORKING SPACE

Derivative and turning points

6 marks

Differentiate

$$\frac{dy}{dx} = 12x^2 + 10x + 2.$$

Solve derivative equal to zero

$$12x^2 + 10x + 2 = 2(3x + 1)(2x + 1) = 0 \Rightarrow x = -\frac{1}{3}, -\frac{1}{2}.$$

Substitute

$$x = -\frac{1}{2} \Rightarrow y = -\frac{1}{4}; \quad x = -\frac{1}{3} \Rightarrow y = -\frac{7}{27}.$$

Final answer

$$\frac{dy}{dx} = 12x^2 + 10x + 2; \quad \left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$$

4MA1-2H Q18 whole

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q18 whole

4 marks

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

Acceleration from a displacement function

4 marks

whole – Acceleration from a displacement function

Differentiate once

$$v = \frac{ds}{dt} = 6t^2 - 10t + 6.$$

Differentiate again

$$a = \frac{dv}{dt} = 12t - 10.$$

Set the required acceleration

$$12t - 10 = 5 \Rightarrow 12t = 15.$$

Solve

$$t = \frac{15}{12} = 1.25 \text{ s.}$$

Final answer

$$t = 1.25 \text{ s}$$

4MA1-1H Q20 M01

4 marks

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

.....

WORKING SPACE

Find the tangent at a minimum

4 marks

M01 – Find the tangent at a minimum**Differentiate**

$$\frac{dy}{dx} = 8x^3 - 64.$$

Set the gradient to zero

$$8x^3 - 64 = 0 \Rightarrow x = 2.$$

Find the y-coordinate

$$y = 2(2^4) - 64(2) = -96.$$

State the horizontal tangent

$$y = -96.$$

Final answer

$$y = -96$$

4MA1-1H Q24

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q24

6 marks

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

Chapter: Differentiation

6 marks

Q24 – Chapter: Differentiation

Family: Find and classify stationary points • Variant: Recover curve parameters from point and gradient data

Method

At $(2, -6)$, $-6 = 8a + 4b - 24 + 6$, so $8a + 4b = 12$.

Why: Substitute the point into the cubic equation.

Method

$\frac{dy}{dx} = 3ax^2 + 2bx - 12$.

Why: Differentiate the curve.

Method

At $x=2$, $16 = 12a + 4b - 12$, so $12a + 4b = 28$.

Why: Use the given gradient at A.

Method

Solving $8a + 4b = 12$ and $12a + 4b = 28$ gives $a = 4$ and $b = -5$.

Why: Solve the simultaneous linear equations.

Method

At $x=3$, $y = 4(3)^3 - 5(3)^2 - 12(3) + 6$.

Why: Substitute $x=3$ and the constants.

Method

$y = 33$.

Why: Evaluate the expression.

Final answer

y: 33

4MA1-2H Q24

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q24

5 marks

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

Chapter: Differentiation

5 marks

Q24 – Chapter: Differentiation

Family: Apply differentiation to linear kinematics • Variant: Instantaneous rest or direction reversal

Method

$$v = dx/dt = 12t^2 - 27.$$

Why: Differentiate $x = 4t^3 - 27t + 8$.

Method

At a reversal $v=0$, so $12t^2 - 27 = 0$ and $t^2 = 9/4$.

Why: Set velocity equal to zero.

Method

Since $t \geq 0$, $t = 3/2$.

Why: Choose the physical time.

Method

$$a = dv/dt = 24t.$$

Why: Differentiate velocity.

Method

At $t = 3/2$, $a = 36 \text{ m/s}^2$.

Why: Evaluate the acceleration.

Final answer

acceleration: 36 m/s^2

23 Curve **C** has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of **C** is negative.

(Total for Question 23 is 4 marks)

Answer reference | May-Nov 2020 | Question 23

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

23	$\left(\frac{dy}{dx} = 3px^2 - m\right)$		4	M1 for $3px^2$ or $-m$
	$3px^2 - m < 0$ oe			M1 ft dep on M1 for setting up an inequality with their ' $3px^2$ ' - ' m ' must be a two-term expression in the form $apx^2 \pm m$
	$\pm \sqrt{\frac{m}{3p}}$			B1 for both critical values
		$-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$		A1 may be seen as two separate inequalities
Total 4 marks				

Worked solution

May-Nov 2020 | Question 23 | 4 marks

Family: Use the derivative as gradient or rate • Variant: Points where gradient equals a value

Method

$$y = px^3 - mx$$

Differentiate:

$$(dy)/(dx) = 3px^2 - m$$

The gradient is negative when

$$3px^2 - m < 0$$

$$3px^2 < m$$

$$x^2 < (m)/(3p)$$

So

$$-\sqrt{(m)/(3p)} < x < \sqrt{(m)/(3p)}$$

$$\text{Answer: } -\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$$

20 The radius of a right circular cylinder is x cm.

The height of the cylinder is $\left(\frac{800}{\pi x} - x\right)$ cm.

The volume of the cylinder is V cm³

Find the maximum value of V

Give your answer correct to the nearest whole number.



(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

20	$(V =) \pi x^2 \left(\frac{800}{\pi x} - x \right)$ or $800x - \pi x^3$ oe		5	M1 for volume of cylinder (in terms of one variable, e.g. x or r)
	$\left(\frac{dV}{dx} = \right) 800 - 3\pi x^2$			M1ft for differentiating an expression in one variable to find 800 or $\pm 3\pi x^2$ (must come from a cubic in the form $800x \pm \pi x^3$ or $800x \pm ax^3$ or $bx \pm \pi x^3$ where $a \neq 0$ and $b \neq 0$)
	" $800 - 3\pi x^2 = 0$ "			M1ft dep on previous M1 for equating their $\frac{dV}{dx}$ to zero (must be a quadratic in the form $800 \pm ax^2$ or $b \pm 3\pi x^2$ where $a \neq 0$ and $b \neq 0$)
	$(x =) \sqrt{\frac{800}{3\pi}} (= \sqrt{84.8(8263...)})$ or $\frac{\sqrt{9600\pi}}{6\pi}$ or $9.2(13177...)$			A1 for a correct value of x Allow use of quadratic formula
	Award marks within the range from correct working	4914		A1 accept 4910 – 4914
				Total 5 marks

Worked solution

January 2023 | Question 20 | 5 marks

Family: Optimise a practical model • Variant: Maximum volume from geometric dimensions

Official final mark-scheme pages are embedded immediately after this question.

4MA1-1H Q23 Q23

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q23 Q23

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n = \dots\dots\dots$

Chapter: Angles in Polygons & Parallel Lines

6 marks

Q23 – Chapter: Angles in Polygons & Parallel Lines

Family: Use regular and general polygon angles • Variant: Solve an irregular polygon angle or side-count problem

Method

The interior-angle sequence has first term 177 and common difference $d = -2$.

Why: Successive listed angles decrease by 2 degrees.

Method

Its sum is $S_n = n/2[2(177) + (n - 1)(-2)]$.

Why: Apply the arithmetic-series sum formula.

Method

The interior-angle sum of an n -sided polygon is $180(n - 2)$, so $n/2[354 - 2(n - 1)] = 180(n - 2)$.

Why: Equate the sequence sum to the polygon angle sum.

Method

Simplifying gives $n^2 + 2n - 360 = 0$.

Why: Expand and collect all terms on one side.

Method

$n^2 + 2n - 360 = (n - 18)(n + 20)$, so $n = 18$ or $n = -20$.

Why: Factorise the quadratic and use the zero-product rule.

Method

The number of sides is positive and $n > 5$, hence $n = 18$.

Why: Apply the domain condition for a polygon.

Final answer

number of sides: $n = 18$

4MA1-1H Q18 demand

5 marks

18 The diagram shows the positions of three villages, A , B and C

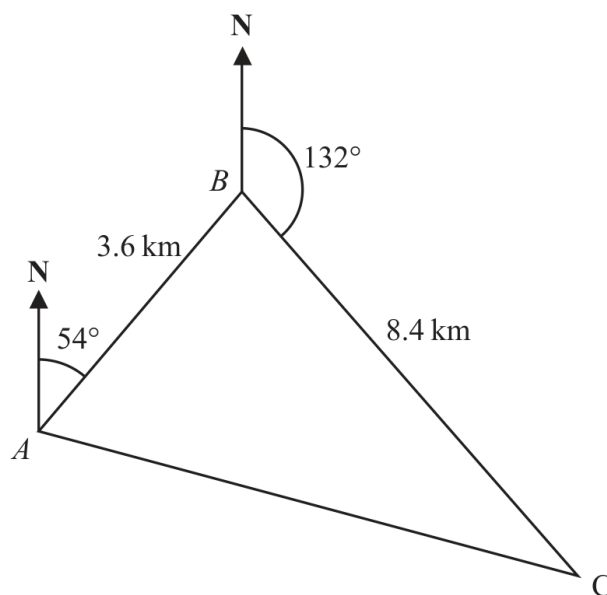


Diagram **NOT**
accurately drawn

The bearing of B from A is 054°

The bearing of C from B is 132°

Melur walks from A to B

She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

WORKING SPACE

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q18 demand

5 marks

18 The diagram shows the positions of three villages, A , B and C

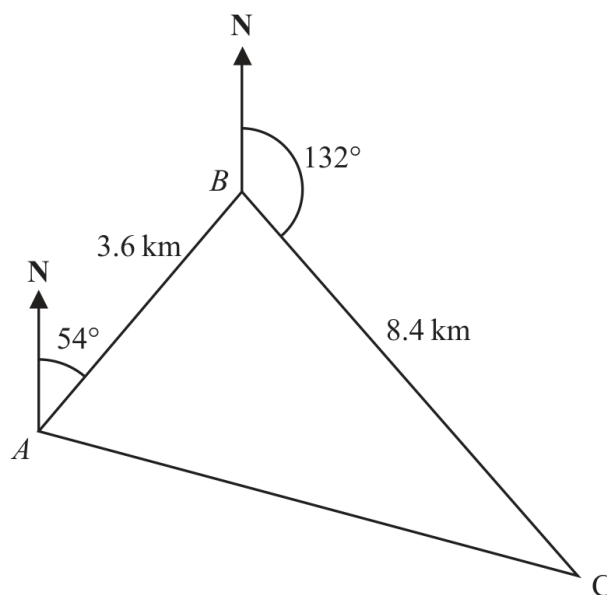


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Melur walks from A to B

She then walks from B to C and from C to A

Melur walks at an average speed of 6 km/h

Work out the total time Melur takes.

Give your answer in hours and minutes.

..... hours minutes

Bearings and total walking time

5 marks

Demand – Bearings and total walking time

Find angle ABC

$$\angle ABC = 54^\circ + (180^\circ - 132^\circ) = 102^\circ.$$

Find AC

$$AC^2 = 3.6^2 + 8.4^2 - 2(3.6)(8.4) \cos 102^\circ \Rightarrow AC \approx 9.802 \text{ km.}$$

Find time

$$t = \frac{3.6 + 8.4 + 9.802}{6} = 3.6337 \text{ h} = 3 \text{ h } 38 \text{ min (approximately).}$$

Final answer

3 hours 38 minutes

4MA1-2H Q24 M01

6 marks

24 A , B and C are three points on horizontal ground.

$$AB = 8.4 \text{ metres} \quad BC = 9.2 \text{ metres}$$

B is on a bearing of 067° from A

C is on a bearing of 129° from B

Calculate the bearing of A from C

Give your answer correct to the nearest degree.

4MA1-2H Q24 M01

6 marks

WORKING SPACE

Find a bearing using the cosine and sine rules

6 marks

M01 – Find a bearing using the cosine and sine rules**Find the included angle**

$$\angle ABC = 67^\circ + (180^\circ - 129^\circ) = 118^\circ.$$

Find AC

$$AC^2 = 8.4^2 + 9.2^2 - 2(8.4)(9.2) \cos 118^\circ \Rightarrow AC = 15.09 \dots$$

Find an angle

Use the sine rule to obtain the angle at C, about 29–30°.

Convert to a bearing

The bearing of A from C is 280° (nearest degree).

Final answer

280 degrees

16 The diagram shows the positions of three ships, A , B and C .

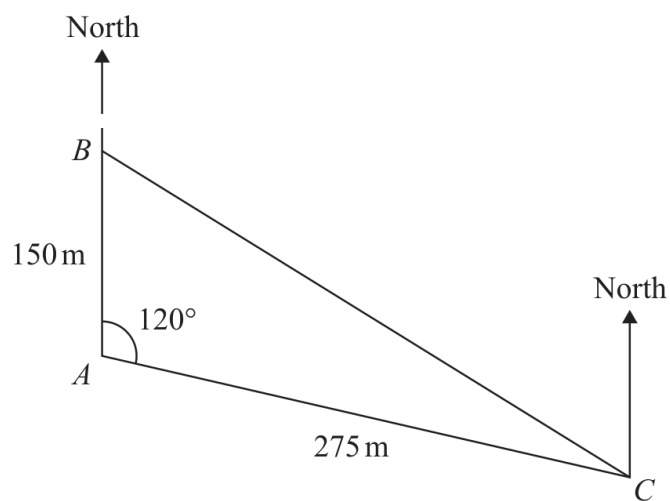


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .

Give your answer correct to the nearest degree.

○

Worked solution

January 2021 | Question 16 | 5 marks

Family: Bearing geometry with trigonometry • Variant: Sine or cosine rule supports a final bearing

Method / working

1. Use the cosine rule: $BC^2 = 150^2 + 275^2 - 2(150)(275)\cos 120^\circ$.

2. $BC = 373.4 \dots$ m.

3. Use the sine rule to find the angle at A or C.

4. The required bearing from B is $120^\circ + 20.3^\circ \sim 140.3^\circ$.

5. To the nearest degree, the bearing is 140° .

Final answer: 140°

4MA1-1H Q22 demand

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

WORKING SPACE

4MA1-1H Q22 demand

6 marks

- 22 [In this question 1 cm = 1 unit on the x -axis and
1 cm = 1 unit on the y -axis]

P is a point on a circle with centre $(0, 0)$

The coordinates of P are $(8, -10)$

The line L is the tangent to the circle at the point P

L crosses the x -axis at the point Q and crosses the y -axis at the point R

Work out the length of RQ

Give your answer correct to 3 significant figures.

..... cm

Length between tangent intercepts

6 marks

Demand – Length between tangent intercepts

Find radius gradient

$$m_{OP} = (-10 - 0)/(8 - 0) = -5/4.$$

Find tangent gradient

$$m_L = 4/5 \text{ since } m_{OP}m_L = -1.$$

Find intercepts

$$y + 10 = \frac{4}{5}(x - 8) \Rightarrow R = (0, -16.4), Q = (20.5, 0).$$

Distance

$$RQ = \sqrt{20.5^2 + 16.4^2} = 26.3 \text{ cm (3 s.f.)}.$$

Final answer

26.3 cm

4MA1-2H Q16 M01, M02, M03, M04

5 marks

16 A, B, C and D are points on a circle, centre O

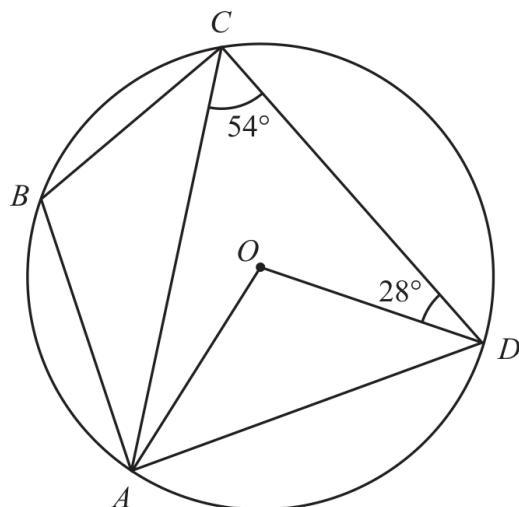


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Continue your method

5 marks

WORKING SPACE

4MA1-2H Q16 M01, M02, M03, M04

5 marks

16 A, B, C and D are points on a circle, centre O

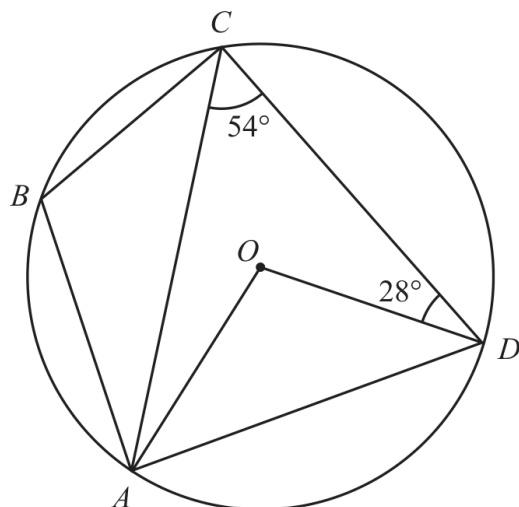


Diagram **NOT**
accurately drawn

(a) (i) Work out the size of angle AOD

.....
(1)

(ii) Give a reason for your answer to part (a)(i)

.....
.....
(1)

(b) Work out the size of angle CAO

.....
(1)

(c) Work out the size of angle ABC

.....
(2)

Find a central angle / Give the circle-theorem reason / Find angle CAO / Find angle ABC

5 marks

M01 – Find a central angle**Use the central-angle theorem**

$$\angle AOD = 2(54^\circ) = 108^\circ.$$

Final answer

108 degrees

M02 – Give the circle-theorem reason**State the theorem**

The angle at the centre is twice the angle at the circumference standing on the same arc.

Final answer

Angle at centre is twice angle at circumference

M03 – Find angle CAO**Use equal radii**OA=OC, so the base angles in triangle AOC are equal; with the central angle, $\angle CAO=26^\circ$.**Final answer**

26 degrees

M04 – Find angle ABC**Find the adjacent angle**

Use the radius/chord angles to obtain the angle at C in triangle ABC.

Use angles in a triangle

$$\angle ABC = 116^\circ.$$

Final answer

116 degrees

4MA1-2H Q19

3 marks

19

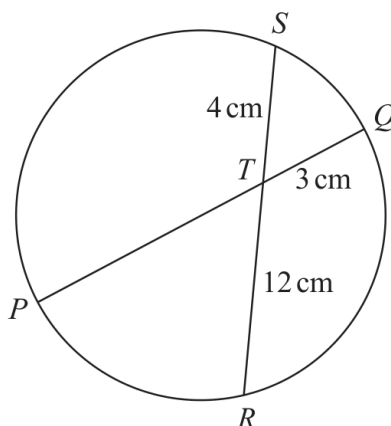


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3\text{ cm}$$

$$ST = 4\text{ cm}$$

$$TR = 12\text{ cm}$$

Calculate the radius of the circle.

..... cm

Continue your method**3 marks**

WORKING SPACE

4MA1-2H Q19

3 marks

19

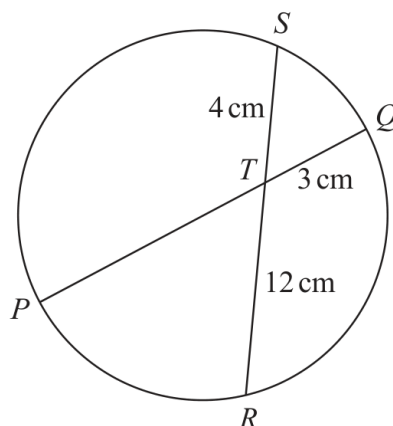


Diagram **NOT** accurately drawn

PTQ is a diameter of a circle.
 RTS is a chord of the circle.

$$TQ = 3\text{ cm} \qquad ST = 4\text{ cm} \qquad TR = 12\text{ cm}$$

Calculate the radius of the circle.

..... cm

Chapter: Circle Theorems

3 marks

Q19 – Chapter: Circle Theorems

Family: Use intersecting chord and secant products • Variant: Use the internal intersecting-chords product

Method

By intersecting chords, $PT \times TQ = ST \times TR$, so $PT \times 3 = 4 \times 12$ and $PT = 16$ cm.

Why: Use the chord-intersection theorem.

Method

$PQ = PT + TQ = 16 + 3 = 19$ cm.

Why: P, T, Q are collinear and PQ is a diameter.

Method

Radius = $PQ/2 = 9.5$ cm.

Why: A radius is half a diameter.

Final answer

radius: 9.5 cm

4MA1-1H Q23

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

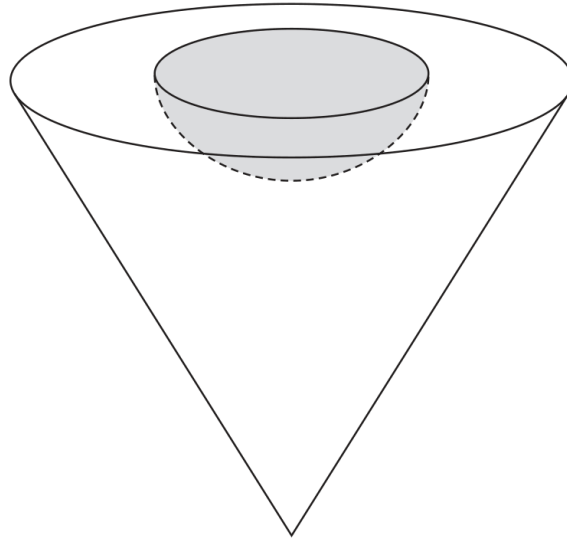


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q23

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

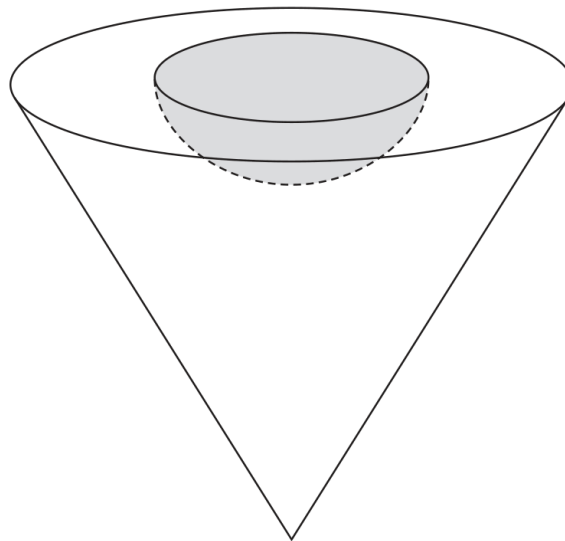


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The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

Cone with a hemisphere removed**5 marks****Find the scale**

$$50\pi x^3 - \frac{16}{3}\pi x^3 = 6948\pi \Rightarrow x^3 = 10422/67.$$

Removed surface

$$3\pi(2x)^2 = 12\pi x^2 = 12\pi(10422/67)^{2/3} = 1090.390\dots$$

Final answer

$$1090 \text{ cm}^2$$

4MA1-1H Q21 demand

5 marks

21 Here is a triangular prism $ABCDEF$

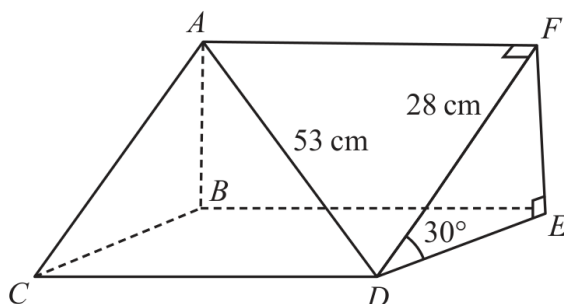


Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DE = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

WORKING SPACE

4MA1-1H Q21 demand

5 marks

21 Here is a triangular prism $ABCDEF$

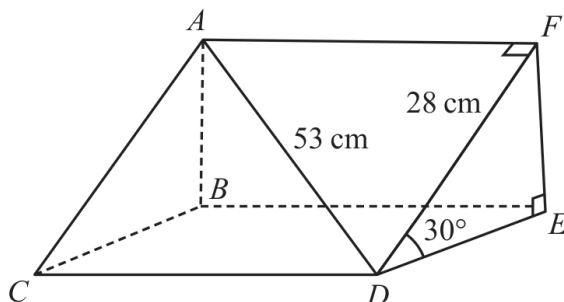


Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

Volume of a triangular prism

5 marks

Demand – Volume of a triangular prism

Resolve the triangular end

$$FE = 28 \sin 30^\circ = 14, \quad DE = 28 \cos 30^\circ = 14\sqrt{3}.$$

Find prism length

$$AF = \sqrt{53^2 - 28^2} = 45.$$

Calculate volume

$$V = \frac{1}{2}(14)(14\sqrt{3})(45) = 4410\sqrt{3} \approx 7638.34.$$

Final answer7638 cm³

4MA1-1H Q22 Q22

5 marks

22 A solid is made from a cone and a hemisphere.

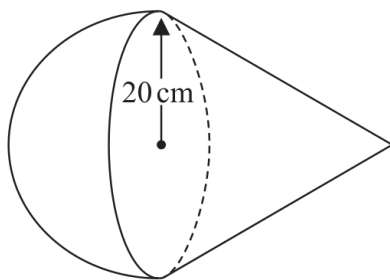


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{cm}^2$

The volume of the solid is $k\pi\text{cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

Continue your method

5 marks

WORKING SPACE

4MA1-1H Q22 Q22

5 marks

22 A solid is made from a cone and a hemisphere.

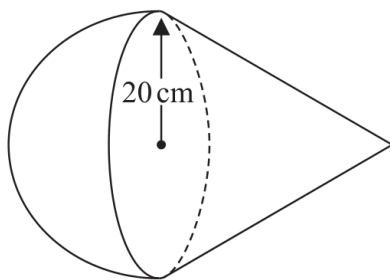


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The volume of the solid is $k\pi\text{ cm}^3$

Work out the exact value of k

$k = \dots\dots\dots$

Chapter: Volume & Surface Area

5 marks

Q22 – Chapter: Volume & Surface Area

Family: Resolve composite solids and volume-unit constraints • Variant: Find volume of a joined composite solid

Method

For the cone, curved surface area $\pi r l = 580\pi$ gives $\pi(20)l = 580\pi$.

Why: Use the given radius in the curved-surface-area formula.

Method

The cone slant height is $l = 29$ cm.

Why: Cancel π and divide by 20.

Method

The cone height is $\sqrt{29^2 - 20^2} = \sqrt{441} = 21$ cm.

Why: The radius, height, and slant height form a right triangle.

Method

The solid volume is $(\frac{2}{3})\pi(20)^3 + (\frac{1}{3})\pi(20)^2(21)$.

Why: Add the hemisphere volume to the cone volume.

Method

This is $(\frac{16000}{3} + \frac{8400}{3})\pi = (\frac{24400}{3})\pi$, so $k = 24400/3$.

Why: Combine the exact coefficients of π .

Final answer

k: $k = 24400/3$

26 Here is a sector, AOB , of a circle with centre O and angle $AOB = x^\circ$

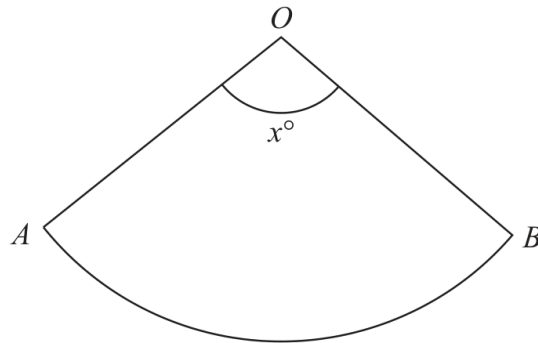


Diagram **NOT**
accurately drawn

The sector can form the curved surface of a cone by joining OA to OB .

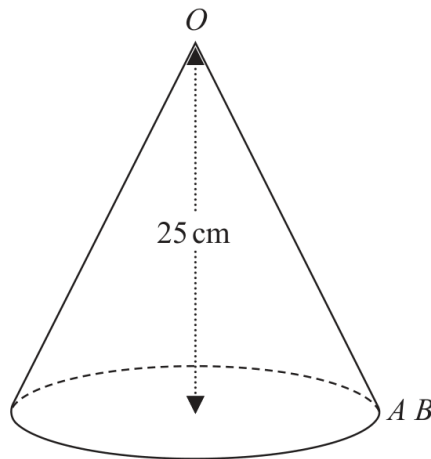


Diagram **NOT**
accurately drawn

The height of the cone is 25 cm .

The volume of the cone is 1600 cm^3

Work out the value of x .

Give your answer correct to the nearest whole number.

$x = \dots\dots\dots$

Worked solution

January 2020 | Question 26 | 6 marks

Family: Use sphere and cone formulae • Variant: Find cone surface area or recover its sector

Method / working

1. Cone volume gives $r^2 = 4800 / (25\pi)$, so $r \sim 7.8176$ cm.
2. Slant height $l = \sqrt{r^2 + 25^2} \sim 26.1938$ cm.
3. Sector arc equals base circumference: $(x/360)2\pi l = 2\pi r$.
4. Cancel 2π and rearrange $x = 360r/l$.
5. $x \sim 107.443^\circ$.
6. Nearest whole degree: $x = 107^\circ$.

Final answer: 107°

17 The diagram shows a solid prism $ABCDEFGH$.

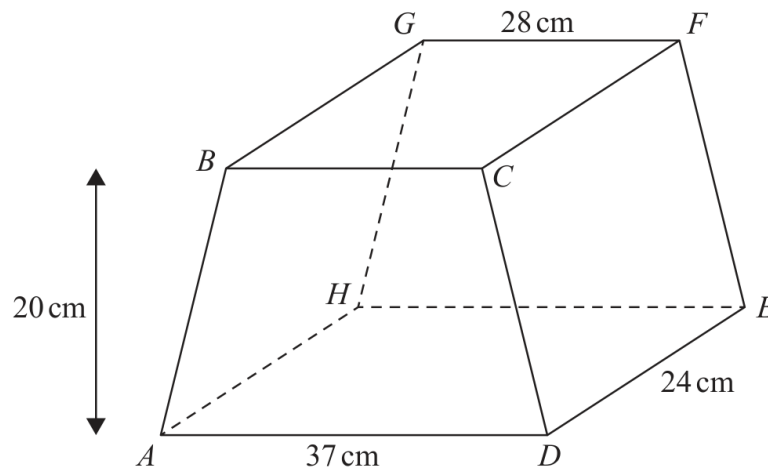


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.

The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

(a) Work out the total surface area of the prism.

..... cm^2
(4)

Worked solution

January 2021 | Question 17 | 4 marks

Family: Find surface area of prisms and cylinders • Variant: Find surface area of a cuboid or simple prism

Method / working

1. Area of trapezium ABCD = $\frac{1}{2}(37+28) \times 20 = 650 \text{ cm}^2$.

2. Sloping side AB = $\sqrt{20^2 + 4.5^2} = 20.5 \text{ cm}$.

3. Add two trapezia and four rectangular faces: $2 \times 650 + 2 \times 20.5 \times 24 + 37 \times 24 + 28 \times 24$.

4. Total surface area = 3844 cm^2 .

Final answer: 3844 cm^2

21 Given that the surface area of a sphere is $49\pi\text{cm}^2$

find the volume of the sphere.

Give your answer correct to the nearest integer.

..... cm^3

(Total for Question 21 is 3 marks)

Answer reference | January 2023 | Question 21

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Question	Working	Answer	Mark	Notes
21	$r = \sqrt{\frac{49\pi}{4\pi}}$ oe (= 3.5)		3	M1
	[volume =] $\frac{4}{3} \times \pi \times 3.5^3$			M1
	<i>Correct answer scores full marks (unless from obvious incorrect working)</i>	180		A1 awrt 180
				Total 3 marks

Worked solution

January 2023 | Question 21 | 3 marks

Family: Use sphere and cone formulae • Variant: Find sphere or hemisphere surface area

Official final mark-scheme pages are embedded immediately after this question.

The height of solid **A** is 31 cm
The height of solid **B** is 18.6 cm

Given that

$$\text{volume of solid A} - \text{volume of solid B} = 735 \text{ cm}^3$$

find the volume of solid **A**

..... cm³

(Total for Question 26 is 4 marks)

WORKING SPACE

Volumes of similar solids

4 marks

Scale volumes

$$A : B = 31 : 18.6 = 5 : 3 \Rightarrow V_A : V_B = 5^3 : 3^3 = 125 : 27.$$

Use the difference

$$V_A = 125k, V_B = 27k; \quad 98k = 735 \Rightarrow k = 7.5.$$

Find A

$$V_A = 125(7.5) = 937.5 \text{ cm}^3.$$

Final answer

$$937.5 \text{ cm}^3 \text{ (or } 938 \text{ cm}^3\text{)}$$

4MA1-1H Q23 demand

5 marks

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

..... cm

WORKING SPACE

4MA1-1H Q23 demand

5 marks

23 Solid A is similar to solid B

Here is some information about solid A and solid B

	solid A	solid B
Height (cm)	3^x	
Area (cm ²)	7776	486
Volume (cm ³)	8^x	2^{x+4}

Work out the height of solid B
Give your answer as a decimal.

..... cm

Similar solids: height of B

5 marks

Demand – Similar solids: height of B

Find the linear scale factor

$$\frac{A_A}{A_B} = \frac{7776}{486} = 16 \Rightarrow k = 4.$$

Use the volume ratio

$$\frac{8^x}{2^{x+4}} = k^3 = 64 \Rightarrow 2^{2x-4} = 2^6 \Rightarrow x = 5.$$

Scale the height

$$h_A = 3^5 = 243, \quad h_B = 243/4 = 60.75 \text{ cm}.$$

Final answer

60.75 cm

4MA1-2H Q23 whole

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

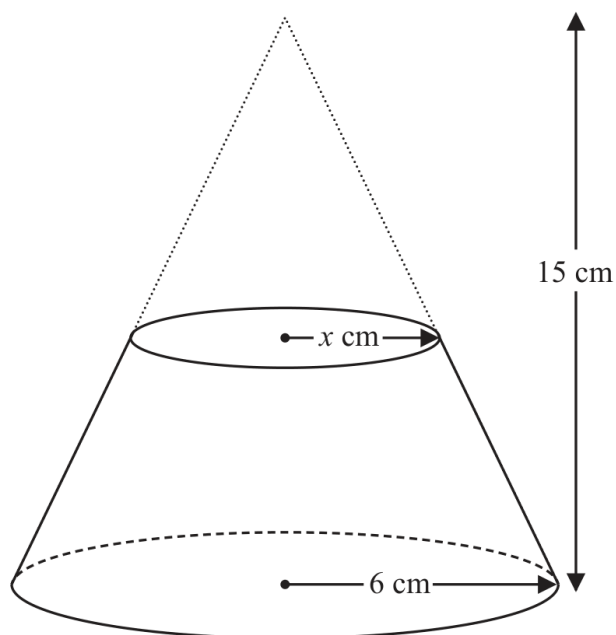


Diagram **NOT**
accurately drawn

The height of the large cone is 15 cm.

The radius of the base of the large cone is 6 cm.

The radius of the base of the small cone is x cm.

Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

WORKING SPACE

Continue your method**5 marks**

WORKING SPACE

4MA1-2H Q23 whole

5 marks

23 Here is a frustum of a cone.

The frustum is made by removing a small cone from a similar large cone.

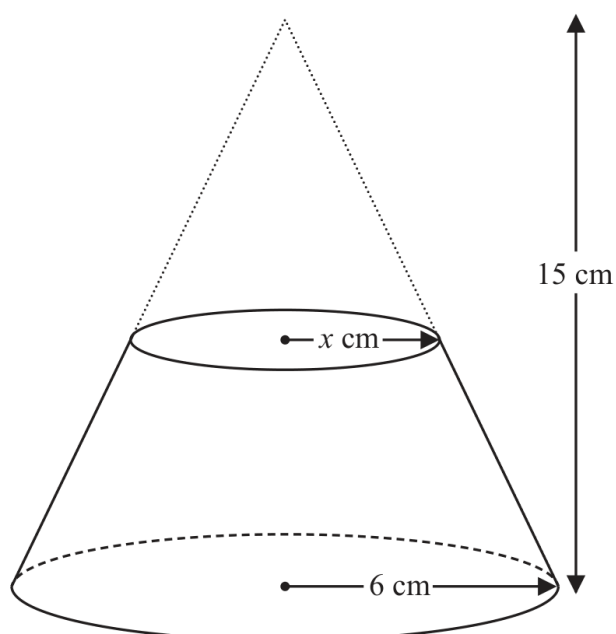


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Given that the volume of the frustum is $\frac{4212}{25}\pi \text{ cm}^3$

work out the value of x

Show clear algebraic working.

$x = \dots\dots\dots$

Radius of a similar-cone frustum

5 marks

whole – Radius of a similar-cone frustum

Find the large-cone volume

$$V_L = \frac{1}{3}\pi(6^2)(15) = 180\pi.$$

Use similarity

The small cone has height $\frac{x}{6}(15) = \frac{5}{2}x$, so its volume is $\frac{1}{3}\pi x^2 \left(\frac{5}{2}x\right) = \frac{5}{6}\pi x^3$.

Use the frustum volume

$$180\pi - \frac{5}{6}\pi x^3 = \frac{4212}{25}\pi \Rightarrow \frac{5}{6}x^3 = \frac{288}{25}.$$

Solve the cubic

$$x^3 = \frac{1728}{125} = \left(\frac{12}{5}\right)^3 \Rightarrow x = \frac{12}{5} = 2.4 \text{ cm}.$$

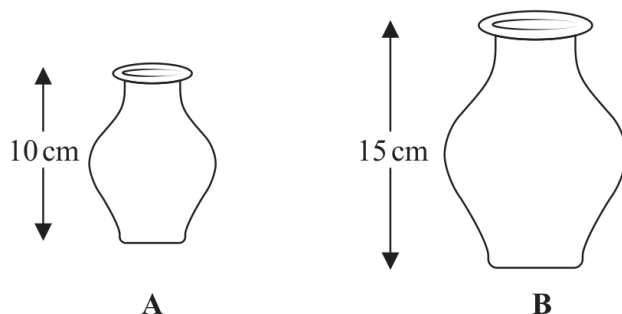
Final answer

$x = 2.4 \text{ cm}$

4MA1-1H Q17

4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

Vase **B** has height 15 cm.

The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

Continue your method

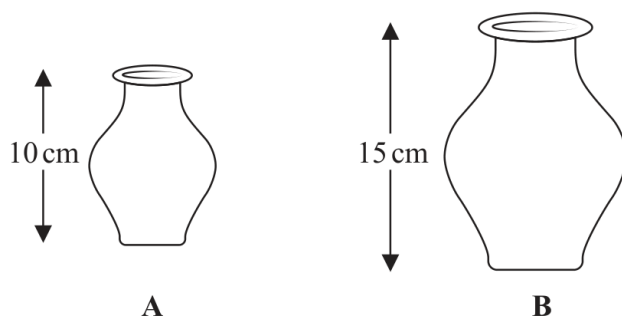
4 marks

WORKING SPACE

4MA1-1H Q17

4 marks

17 **A** and **B** are two similar vases.



Vase **A** has height 10 cm.

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The difference between the volume of vase **A** and the volume of vase **B** is 1197 cm^3

Calculate the volume of vase **A**

..... cm^3

Chapter: Area & Volume of Similar Shapes

4 marks

Q17 – Chapter: Area & Volume of Similar Shapes

Family: Use volume scale factors of similar solids • Variant: Solve a volume difference or sum for similar solids

Method

The linear scale factor from A to B is $15/10=1.5$, so the volume scale factor is $1.5^3=27/8$.

Why: Similar solids scale volumes by the cube of the linear factor.

Method

If V is the volume of A, $(27/8)V - V = 1197$.

Why: Use the stated volume difference.

Method

$(19/8)V = 1197$, so $V = 1197 \times 8/19$.

Why: Solve the linear equation in V .

Method

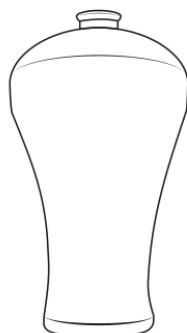
$V = 504 \text{ cm}^3$.

Why: Evaluate and state the volume of vase A.

Final answer

volume of A: 504 cm^3

19 **A** and **B** are two similar vases.



A



B

Diagram **NOT**
accurately drawn

The vases are such that

$$\text{surface area of vase } \mathbf{B} = \frac{25}{64} \times \text{surface area of vase } \mathbf{A}$$

and that

$$\text{volume of vase } \mathbf{A} - \text{volume of vase } \mathbf{B} = 541.8 \text{ cm}^3$$

Calculate the volume of vase **B**

..... cm³

(Total for Question 19 is 4 marks)

Answer reference | January 2023 | Question 19

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Question	working	Answer	Mark	Notes
19	$\text{eg } \sqrt{\frac{25}{64}} \left(= \frac{5}{8} = 0.625 \right) \text{ or } \sqrt{\frac{64}{25}} \left(= \frac{8}{5} = 1.6 \right) \text{ or}$ $\sqrt{25} : \sqrt{64} \text{ (5:8) or } \sqrt{64} : \sqrt{25} \text{ (8:5) or}$ $\frac{(\sqrt{25})^3}{(\sqrt{64})^3} = \left(\frac{125}{512} = 0.244140625 \right) \text{ oe or } \frac{512}{125} = 4.096$ $\frac{25^3}{64^3} = \frac{(\text{vol of B})^2}{(\text{vol of B} + 541.8)^2} \text{ or } \frac{25}{64} = \frac{(\text{vol of B})^{\frac{2}{3}}}{(\text{vol of B} + 541.8)^{\frac{2}{3}}} \text{ oe}$		4	M1 for a correct scale factor for length – may be given as a fraction or decimal or ratio or a correct scale factor for volume given as a fraction or decimal or ratio or a correct equation for the volume of vase B
	$\text{eg B} \left(\frac{512}{125} - 1 \right) = 541.8 \text{ or } 3.096\text{B} = 541.8 \text{ oe}$ $\text{or eg A} \left(1 - \frac{125}{512} \right) = 541.8 \text{ or } \frac{387}{512} \text{A} = 541.8$ $(8^3)k - (5^3)k (= 387k) = 541.8$ $\text{or eg } (k =) \frac{541.8}{387} \left(= \frac{7}{5} \right)$			M1 For a correct equation for the volume of B or a correct equation for the volume of A
	$\text{eg (B)} 541.8 \div \frac{387}{125} \text{ or } 541.8 \div 3.096 \text{ or eg } 125 \times \frac{7}{5} \text{ or}$ $\text{(A)} 541.8 \div \frac{387}{512} (= 716.8) \text{ oe}$			M1 For a completely correct method to find the volume of vase B or vase A
	Correct answer scores full marks (unless from obvious incorrect working)	175		A1 cao
Total 4 marks				

Worked solution

January 2023 | Question 19 | 4 marks

Family: Use area scale factors of similar figures • Variant: Solve an area difference or sum for similar figures

Official final mark-scheme pages are embedded immediately after this question.

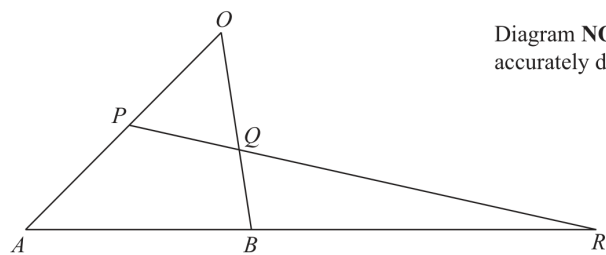


Diagram **NOT**
accurately drawn

OAB is a triangle.
 P is the midpoint of OA
 Q is a point on OB

ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

(1)

WORKING SPACE

4MA1-1H Q23

23

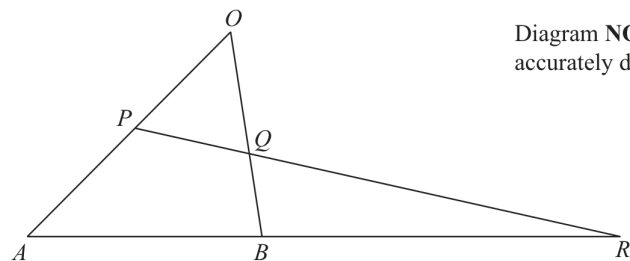


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$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

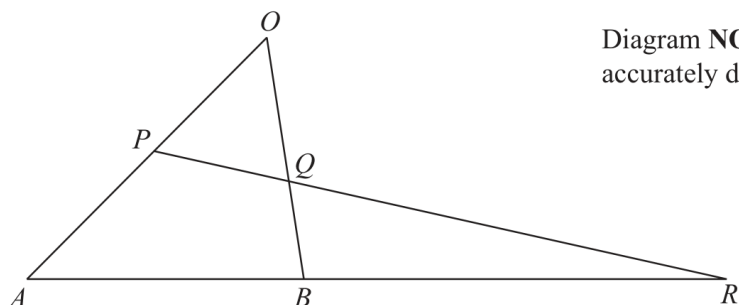
(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

WORKING SPACE

4MA1-1H Q23

23

Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \dots\dots\dots (4)$$

Vector method on two straight lines

5 marks

Find AB

$$\vec{AB} = \vec{OB} - \vec{OA} = 8\mathbf{b} - 12\mathbf{a}.$$

Find PR

$$\vec{OR} = \vec{OA} + 3\vec{AB} = -24\mathbf{a} + 24\mathbf{b}, \quad \vec{OP} = 6\mathbf{a}, \quad \vec{PR} = -30\mathbf{a} + 24\mathbf{b}.$$

Use collinearity

$$\vec{OQ} = \vec{OP} + \lambda\vec{PR} = 6\mathbf{a} + \lambda(-30\mathbf{a} + 24\mathbf{b}). \quad 6 - 30\lambda = 0 \Rightarrow \lambda = \frac{1}{5}, \quad n = 24\lambda = \frac{24}{5}.$$

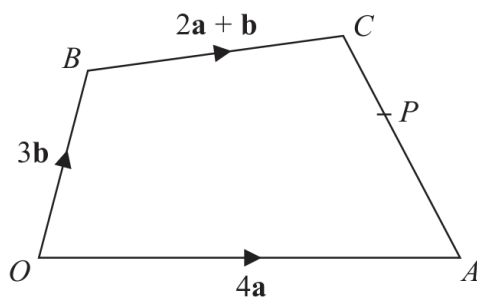
Final answer

$$n = \frac{24}{5}$$

4MA1-2H Q24

6 marks

24

Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

WORKING SPACE

4MA1-2H Q24

6 marks

24

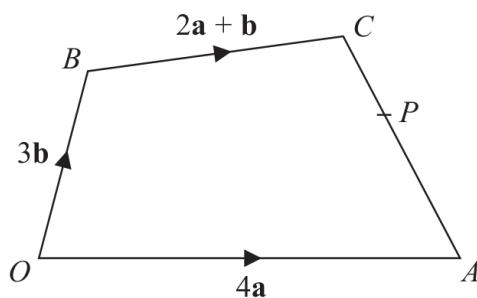


Diagram **NOT** accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
 Give your answer in its simplest form.
 Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

Continue your method**6 marks**

WORKING SPACE

4MA1-2H Q24

6 marks

24

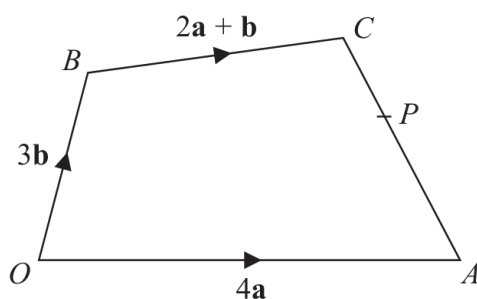


Diagram **NOT**
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- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

4MA1-2H Q24

6 marks

24

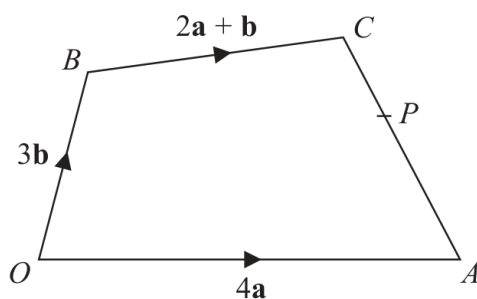


Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

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The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\vec{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\vec{OQ} = \dots\dots\dots (4)$$

Vectors and collinearity

6 marks

Part (a)

$$\vec{AC} = 4b - 2a.$$

Point P

$$\vec{OP} = 4a + \frac{3}{5}(4b - 2a) = \frac{14}{5}a + \frac{12}{5}b.$$

Point Q

$$\vec{OQ} = \lambda \vec{OP} = 3b + \mu(2a + b) \Rightarrow \lambda = 3.$$

Final answer

$$\vec{OQ} = \frac{42}{5}a + \frac{36}{5}b$$

4MA1-2H Q22 (a)(i), (a)(ii), (b)

5 marks

22

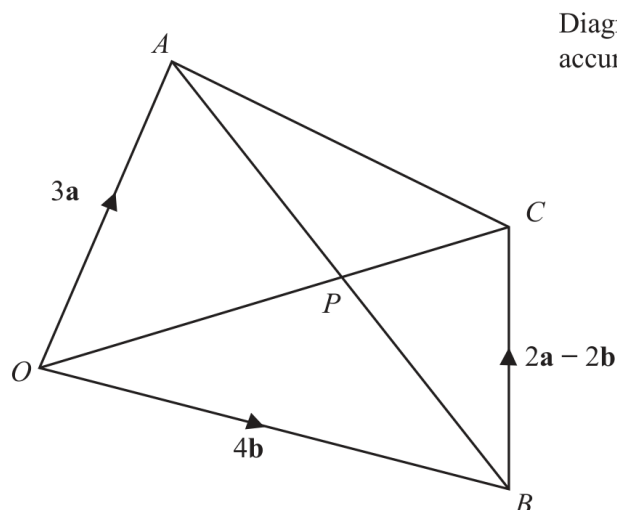


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots (1)$$

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots (1)$$

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots (3)$$

Continue your method

5 marks

WORKING SPACE

4MA1-2H Q22 (a)(i), (a)(ii), (b)

5 marks

22

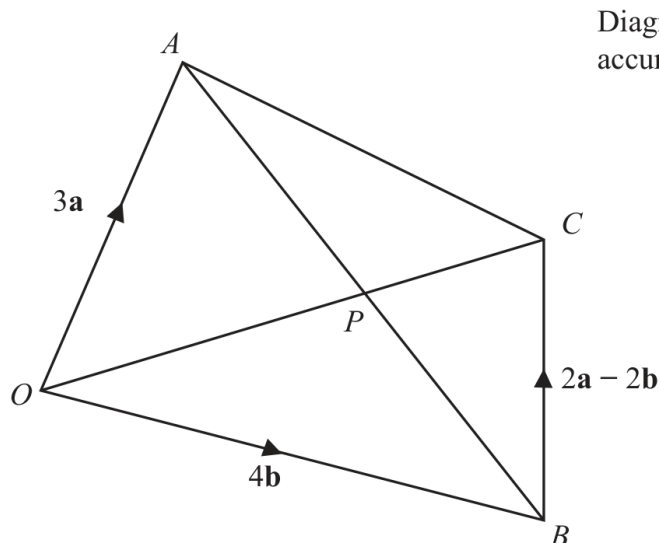


Diagram **NOT**
accurately drawn

$OACB$ is a quadrilateral.

$$\vec{OA} = 3\mathbf{a} \quad \vec{OB} = 4\mathbf{b} \quad \vec{BC} = 2\mathbf{a} - 2\mathbf{b}$$

- (a) (i) Find the vector $\vec{OC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answer.

$$\vec{OC} = \dots\dots\dots$$

(1)

- (ii) Find the vector $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$

$$\vec{AB} = \dots\dots\dots$$

(1)

The point P lies on AB and on OC

- (b) Using a vector method, find the ratio $AP : PB$
Show your working clearly.

$$\dots\dots\dots$$

(3)

Find OC from two vectors / Find AB from position vectors / Ratio where two vector descriptions meet **5 marks**

(a)(i) – Find OC from two vectors

Use the path O-B-C

$$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 4b + (2a - 2b) = 2a + 2b.$$

Final answer

$$OC = 2a + 2b$$

(a)(ii) – Find AB from position vectors

Subtract position vectors

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 4b - 3a.$$

Final answer

$$AB = 4b - 3a$$

(b) – Ratio where two vector descriptions meet

Describe P on both lines

$$\overrightarrow{OP} = \lambda(2a + 2b) = 3a + \mu(4b - 3a).$$

Compare coefficients

$$2\lambda = 4\mu, \quad 2\lambda = 3 - 3\mu \Rightarrow \mu = \frac{3}{7}.$$

Convert the parameter to a ratio

$$AP : PB = \mu : (1 - \mu) = \frac{3}{7} : \frac{4}{7} = 3 : 4.$$

Final answer

$$AP:PB = 3:4$$

4MA1-1H Q19

5 marks

19

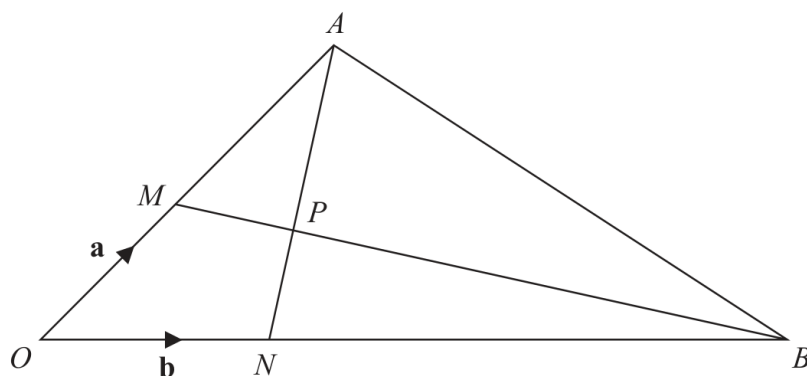


Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....
(1)

(b) Use a vector method to find the ratio $AP:PN$

$AP:PN =$
(4)

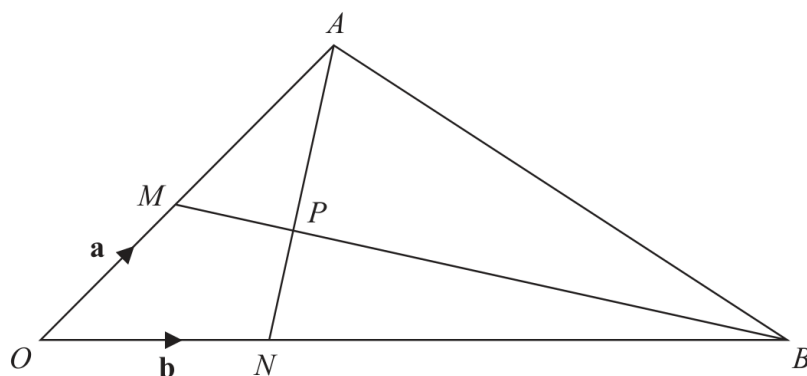
Continue your method**5 marks**

WORKING SPACE

4MA1-1H Q19

5 marks

19

Diagram **NOT**
accurately drawn

OMA , ONB , MPB and NPA are straight lines.

M is the midpoint of OA

$ON:NB = 1:5$

$$\vec{OM} = \mathbf{a} \quad \vec{ON} = \mathbf{b}$$

(a) Find in terms of $\mathbf{a}$ and $\mathbf{b}$ the vector $\vec{AN}$

.....
(1)

(b) Use a vector method to find the ratio $AP:PN$

$AP:PN =$
(4)

Intersection ratio by vectors

5 marks

Write the key vectors

$$\overrightarrow{OA} = 2\mathbf{a}, \quad \overrightarrow{OB} = 6\mathbf{b}, \quad \overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}.$$

Describe P along AN

$$\overrightarrow{OP} = 2\mathbf{a} + \lambda(\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + \lambda\mathbf{b}.$$

Describe P along MB

$$\overrightarrow{OP} = \mathbf{a} + \mu(6\mathbf{b} - \mathbf{a}) = (1 - \mu)\mathbf{a} + 6\mu\mathbf{b}.$$

Equate coefficients

$$\lambda = 6\mu, \quad 2 - 2\lambda = 1 - \mu \Rightarrow \lambda = \frac{6}{11}.$$

Final answer

$$\overrightarrow{AN} = \mathbf{b} - 2\mathbf{a}; \quad AP : PN = 6 : 5$$

4MA1-1H Q24 M01, M02, M03

6 marks

24 OAB is a triangle.

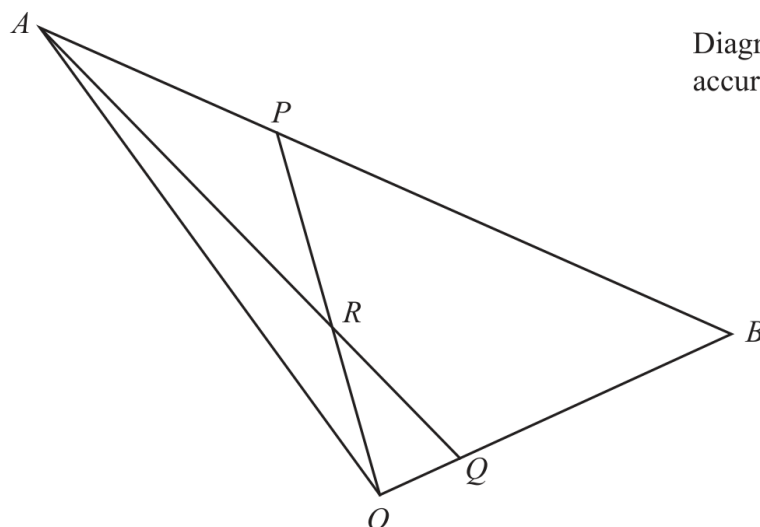


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\vec{OR}$
(4)

WORKING SPACE

4MA1-1H Q24 M01, M02, M03

6 marks

24 OAB is a triangle.

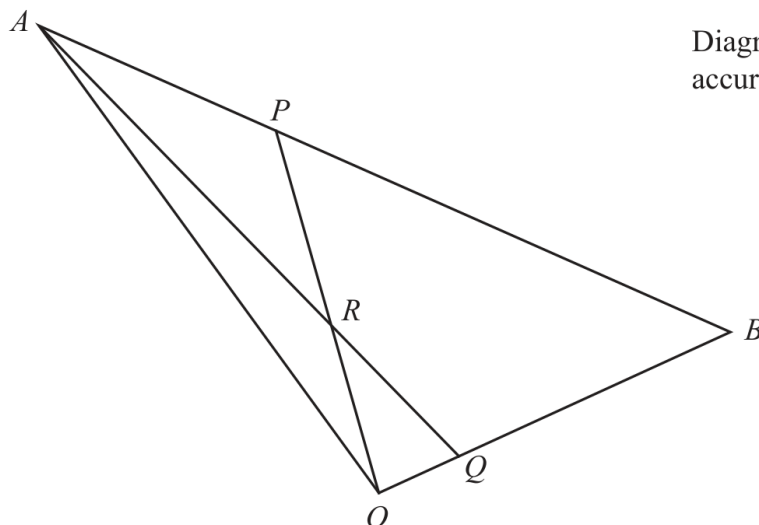


Diagram **NOT**
accurately drawn

$$\vec{OA} = 10\mathbf{a} \quad \vec{OB} = 10\mathbf{b}$$

ARQ and ORP are straight lines.

$$\vec{AP} = \frac{1}{4} \vec{AB} \quad \text{and} \quad \vec{OQ} = \frac{1}{5} \vec{OB}$$

Write the following vectors in terms of $\mathbf{a}$ and $\mathbf{b}$
Simplify your answers.

(i) $\vec{AQ}$

.....
(1)

(ii) $\vec{OP}$

.....
(1)

4MA1-1H Q24 M01, M02, M03

6 marks

(iii) $\vec{OR}$
(4)

Find AQ / Find OP / Find OR from two straight-line routes

6 marks

M01 – Find AQ

Use a route through O

$$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ} = -10\mathbf{a} + 2\mathbf{b}.$$

Final answer

$$-10\mathbf{a} + 2\mathbf{b}$$

M02 – Find OP

Use AP=AB/4

$$\overrightarrow{AP} = \frac{1}{4}(10\mathbf{b} - 10\mathbf{a}) = \frac{5}{2}\mathbf{b} - \frac{5}{2}\mathbf{a}.$$

Add OA

$$\overrightarrow{OP} = 10\mathbf{a} + \overrightarrow{AP} = \frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}.$$

Final answer

$$\frac{15}{2}\mathbf{a} + \frac{5}{2}\mathbf{b}$$

M03 – Find OR from two straight-line routes

Write OR through P

$$\overrightarrow{OR} = \overrightarrow{OP} + \lambda\overrightarrow{PR}.$$

Write OR through Q

Use the line ARQ and the known AQ route to form a second expression.

Equate coefficients

The two routes give $\lambda=1/2$ and hence vector $\overrightarrow{OR}=(15)/(4)\mathbf{a}+5/4\mathbf{b}$.

Simplify

$$\overrightarrow{OR} = \frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}.$$

$$\frac{15}{4}\mathbf{a} + \frac{5}{4}\mathbf{b}$$

4MA1-2H Q17 M01

3 marks

17 Here are two vectors.

$$\vec{FG} = \begin{pmatrix} -5 \\ 2 \end{pmatrix} \quad \vec{HG} = \begin{pmatrix} 4 \\ 14 \end{pmatrix}$$

Calculate the magnitude of the vector $\vec{HF}$

.....

WORKING SPACE

Find the magnitude of HF

3 marks

M01 – Find the magnitude of HF**Find the vector**

$$\overrightarrow{HF} = \overrightarrow{HG} - \overrightarrow{FG} = (4, 14) - (-5, 2) = (9, 12).$$

Use Pythagoras

$$|HF| = \sqrt{9^2 + 12^2} = 15.$$

State the magnitude

$$|HF| = 15.$$

Final answer

15

19 OAB is a triangle.

$$\vec{OA} = \mathbf{a} \quad \vec{OB} = \mathbf{b}$$

The point C lies on OA such that $OC : CA = 1 : 2$

The point D lies on OB such that $OD : DB = 1 : 2$

Using a vector method, prove that $ABDC$ is a trapezium.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

19	$\overrightarrow{AB} = -\mathbf{a} + \mathbf{b}$ or $\overrightarrow{BA} = \mathbf{a} - \mathbf{b}$			M1	Correct diagram (condone missing vector labels or arrows – with C on line segment OA and D on line segment OB) OR for finding $\overrightarrow{AB}$ or $\overrightarrow{BA}$ - may be seen as part of later working
	$\overrightarrow{CD} = \frac{1}{3}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{DC} = \frac{1}{3}(\mathbf{a} - \mathbf{b})$ oe			M1	Method to find $\overrightarrow{CD}$ or $\overrightarrow{DC}$
		Correct vectors and conclusion including <u>parallel</u> and <u>trapezium</u>	3	A1	eg $\overrightarrow{AB}$ (AB) and $\overrightarrow{CD}$ (CD) are parallel therefore $ABDC$ is a trapezium
					Total 3 marks

Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Use vectors with ratios and intersections • Variant: Use vectors in a parallelogram, trapezium or polygon path

Official final mark-scheme pages are embedded immediately after this question.

24 $OAED$ is a quadrilateral.

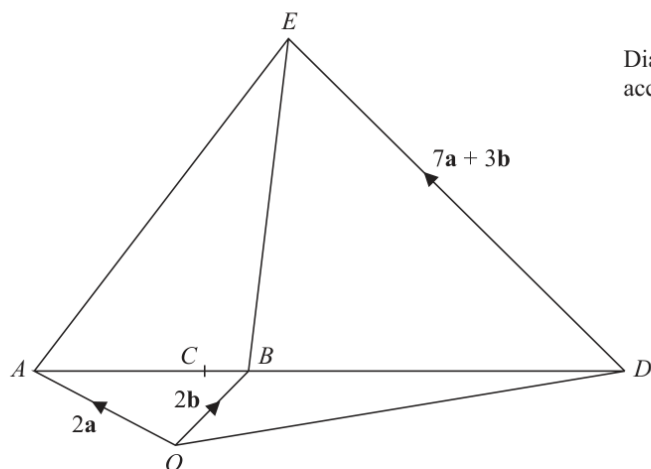


Diagram **NOT**
accurately drawn

$$\vec{OA} = 2\mathbf{a} \quad \vec{OB} = 2\mathbf{b} \quad \vec{DE} = 7\mathbf{a} + 3\mathbf{b}$$

$$AB : BD = 1 : 2$$

The point C on AB is such that OCE is a straight line.

Use a vector method to find the ratio of $OC : CE$



(Total for Question 24 is 5 marks)

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe or $(\overrightarrow{OC}) = 2\mathbf{a} + \lambda(2\mathbf{b} - 2\mathbf{a}) = (2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b}$ oe or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b})$ or $(\overrightarrow{CE}) = (2\mathbf{b} - 2\mathbf{a}) - \lambda(2\mathbf{b} - 2\mathbf{a}) + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} = (1 + 2\lambda)\mathbf{a} + (9 - 2\lambda)\mathbf{b}$			M2 for 2 correct paths seen M1 for 1 correct path seen Any correct path for OC passing through A or B involving a variable
	e.g. $\frac{2 - 2\lambda}{2\lambda} = \frac{1 + 2\lambda}{9 - 2\lambda}$ oe or $\frac{2 - 2\lambda}{2\lambda} = \frac{3}{9}$ oe or $\frac{(1 + 2\lambda)}{(9 - 2\lambda)} = \frac{1}{3}$ oe or $\lambda = \frac{3}{4}$ or $(2 - 2\lambda)\mathbf{a} + 2\lambda\mathbf{b} = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{3}{4}$ or $\mu = \frac{1}{6}$ or $2\mathbf{b} + \lambda(2\mathbf{a} - 2\mathbf{b}) = \mu(3\mathbf{a} + 9\mathbf{b})$ or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1 for comparing coefficients of $\mathbf{a}$ and $\mathbf{b}$ for $(OC \text{ and } CE)$ or $(OC \text{ and } OE)$ or $(CE \text{ and } OE)$ OC is a multiple of OE Two different paths for OC
		1 : 5		A1 dep M2 oe e.g 2 : 10
	Working required			Total 5 marks

Answer reference | January 2023 | Question 24

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

24 ALT	e.g. $(\overrightarrow{AB}) = 2\mathbf{b} - 2\mathbf{a}$ oe or $(\overrightarrow{BA}) = 2\mathbf{a} - 2\mathbf{b}$ oe or $(\overrightarrow{BD}) = 2(2\mathbf{b} - 2\mathbf{a}) (= 4\mathbf{b} - 4\mathbf{a})$ oe or $(\overrightarrow{AD}) = 3(2\mathbf{b} - 2\mathbf{a}) (= 6\mathbf{b} - 6\mathbf{a})$ oe		5	M1
	e.g. $(\overrightarrow{OE}) = 2\mathbf{b} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= 3\mathbf{a} + 9\mathbf{b})$ oe			M1
	e.g. $(\overrightarrow{AE}) = 2\mathbf{b} - 2\mathbf{a} + 2(2\mathbf{b} - 2\mathbf{a}) + 7\mathbf{a} + 3\mathbf{b} (= \mathbf{a} + 9\mathbf{b})$ oe			M1
	$[\overrightarrow{AE} = \lambda \overrightarrow{AD} + \mu \overrightarrow{OE}]$ $\mathbf{a} + 9\mathbf{b} = \lambda(6\mathbf{b} - 6\mathbf{a}) + \mu(3\mathbf{a} + 9\mathbf{b})$ oe or $1 = -6\lambda + 3\mu$ oe and $9 = 6\lambda + 9\mu$ oe or or $\lambda = \frac{1}{4}$ or $\mu = \frac{5}{6}$			M1
		1 : 5		A1 dep on M2 oe e.g 2 : 10
	Working required			Total 5 marks

Worked solution

January 2023 | Question 24 | 5 marks

Family: Use vectors with ratios and intersections • Variant: Find a section or midpoint vector

Official final mark-scheme pages are embedded immediately after this question.